ES827 – ROBÓTICA INDUSTRIAL

Profª Ludmila Corrêa de Alkmin e Silva (ludmila@fem.unicamp.br)

EQUAÇÃO DE MOVIMENTO

$$D(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = \tau$$

$$\begin{bmatrix} d_{11} & d_{12} & d_{13} \\ d_{21} & d_{22} & d_{23} \\ d_{31} & d_{32} & d_{33} \end{bmatrix} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \\ \ddot{d}_3 \end{Bmatrix} + \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} \begin{Bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{d}_3 \end{Bmatrix}$$

$$+ \begin{bmatrix} 0 \\ -gS_2(d_2 m_2 + d_2 m_3 + d_3 m_3) \\ g m_3 C_2 \end{bmatrix} = \begin{Bmatrix} \tau_1 \\ \tau_2 \\ F_3 \end{Bmatrix}$$

EQUAÇÃO DE MOVIMENTO

$$D(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = \tau$$

Matriz de Inércia $D = m[J_v]^T[J_v] + [J_w]^T[R][I][R]^T[J_w]$

Símbolo de Christoffel $C = [C_1]\dot{q}_1 + [C_2]\dot{q}_2 + \cdots + [C_n]\dot{q}_n$

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

Energia potencial P

$$g = \begin{bmatrix} \frac{\partial P}{\partial q_1} \\ \frac{\partial P}{\partial q_2} \\ \vdots \\ \frac{\partial P}{\partial q_n} \end{bmatrix}$$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

Simplicações:

- Inverter os 2 primeiros dígitos gera resultados iguais

$C_{\alpha\beta\gamma} = C_{\beta\alpha\gamma}$ Ex: $C_{311} = C_{131}$

$$C_{232} = C_{322}$$

- Inverter os 2 últimos dígitos inverte o sinal do resultado

$C_{\alpha\beta\gamma} = -C_{\alpha\gamma\beta}$ Ex: $C_{223} = -C_{232}$

$$C_{112} = -C_{121}$$

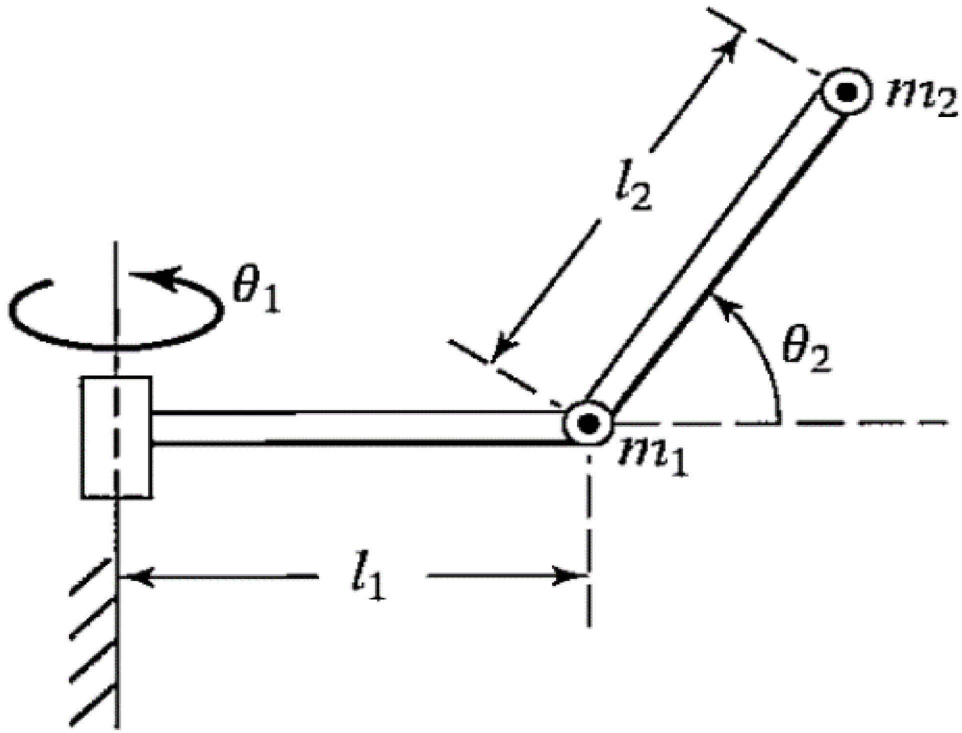
EXEMPLO

Determine a dinâmica do manipulador mostrado abaixo, considerando que as massas m_1 e m_2 dos braços está concentrada nas extremidade do respectivo braço. Considere os tensores de inércia mostrados abaixo.

$$I_1 = \begin{bmatrix} I_{xx1} & 0 & 0 \\ 0 & I_{yy1} & 0 \\ 0 & 0 & I_{zz1} \end{bmatrix}$$

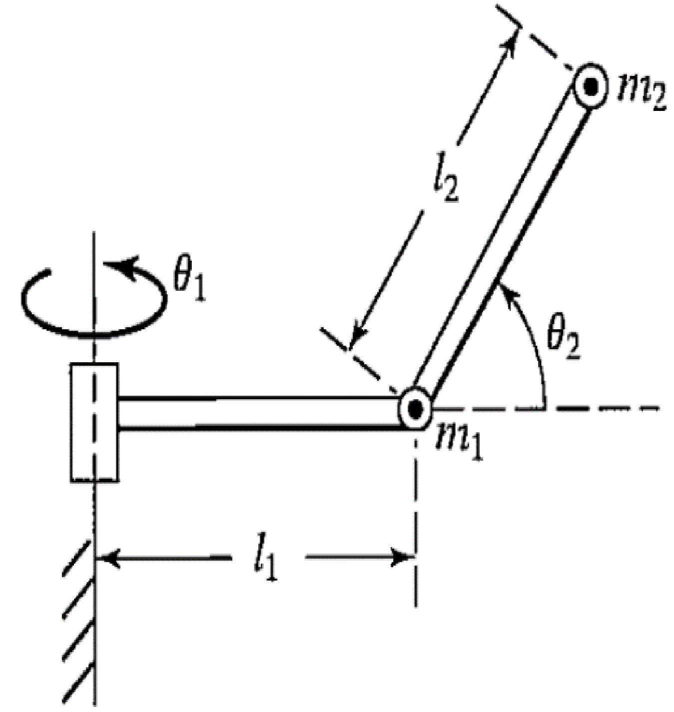
$$I_2 = \begin{bmatrix} I_{xx2} & 0 & 0 \\ 0 & I_{yy2} & 0 \\ 0 & 0 & I_{zz2} \end{bmatrix}$$

Elo	a_i	α_i	d_i	θ_i
1	l_1	90	*0*	θ_1
2	l_2	0	0	θ_2



CINEMÁTICA DIRETA

Elo	a_i	α_i	d_i	θ_i
1	l_1	90	0	θ_1
2	l_2	0	0	θ_2



$$A_1^0 = \begin{bmatrix} C_1 & 0 & S_1 & l_1C_1 \\ S_1 & 0 & -C_1 & l_1S_1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad A_2^1 = \begin{bmatrix} C_2 & -S_2 & 0 & l_2C_2 \\ S_2 & C_2 & 0 & l_2S_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2^0 = \begin{bmatrix} C_1C_2 & -C_1S_2 & S_1 & l_1C_1 + l_2C_1C_2 \\ C_1S_1 & -S_1S_2 & -C_1 & l_1S_1 + l_2C_2S_1 \\ S_2 & C_2 & 0 & l_2S_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$O_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad O_1 = \begin{bmatrix} l_1C_1 \\ l_1S_1 \\ 0 \end{bmatrix} \quad O_2 = \begin{bmatrix} l_1C_1 + l_2C_1C_2 \\ l_1S_1 + l_2C_2S_1 \\ l_2S_2 \end{bmatrix} Z_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} Z_1 = Z_2 = \begin{bmatrix} S_1 \\ -C_1 \\ 0 \end{bmatrix}$$

JACOBIANO

$$O_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad O_1 = \begin{bmatrix} l_1 C_1 \\ l_1 S_1 \\ 0 \end{bmatrix} \quad O_2 = \begin{bmatrix} l_1 C_1 + l_2 C_1 C_2 \\ l_1 S_1 + l_2 C_2 S_1 \\ l_2 S_2 \end{bmatrix} Z_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \mathrm{Z}_1 = \mathrm{Z}_2 = \begin{bmatrix} S_1 \\ -C_1 \\ 0 \end{bmatrix}$$

$$J_1 = \begin{bmatrix} Z_0 \times (O_1 - O_0) & 0 \\ Z_0 & 0 \end{bmatrix} = \begin{bmatrix} -l_1 S_1 & 0 \\ l_1 C_1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$J_2 = \begin{bmatrix} Z_0 \times (O_2 - O_0) & Z_1 \times (O_2 - O_1) \\ Z_0 & Z_1 \end{bmatrix} = \begin{bmatrix} -l_1 S_1 - l_2 S_1 C_2 & -l_2 C_1 S_2 \\ l_1 C_1 + l_2 C_1 C_2 & -l_2 S_1 S_2 \\ 0 & l_2 C_2 \\ 0 & S_1 \\ 0 & -C_1 \\ 1 & 0 \end{bmatrix}$$

PARA O ELO 1

$$J_1 = \begin{bmatrix} Z_0 \times (O_1 - O_0) & 0 \\ Z_0 & 0 \end{bmatrix} = \begin{bmatrix} -l_1 S_1 & 0 \\ l_1 C_1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$J_{v1} = \begin{bmatrix} -l_1 S_1 & 0 \\ l_1 C_1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$J_{w1} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$A_1^0 = \begin{bmatrix} C_1 & 0 & S_1 & l_1 C_1 \\ S_1 & 0 & -C_1 & l_1 S_1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \longrightarrow R_1 = \begin{bmatrix} C_1 & 0 & S_1 \\ S_1 & 0 & -C_1 \\ 0 & 1 & 0 \end{bmatrix}$$

PARA O ELO 2

$$J_2 = \begin{bmatrix} Z_0 \times (O_2 - O_0) & Z_1 \times (O_2 - O_1) \\ Z_0 & Z_1 \end{bmatrix}$$

$$= \begin{bmatrix} -l_1S_1 - l_2S_1C_2 & -l_2C_1S_2 \\ l_1C_1 + l_2C_1C_2 & -l_2S_1S_2 \\ 0 & l_2C_2 \\ 0 & S_1 \\ 0 & -C_1 \\ 1 & 0 \end{bmatrix}$$

$$J_{v2} = \begin{bmatrix} -l_1S_1 - l_2S_1C_2 & -l_2C_1S_2 \\ l_1C_1 + l_2C_1C_2 & -l_2S_1S_2 \\ 0 & l_2C_2 \end{bmatrix} \qquad J_{w2} = \begin{bmatrix} 0 & S_1 \\ 0 & -C_1 \\ 1 & 0 \end{bmatrix}$$

$$A_2^0 = \begin{bmatrix} C_1C_2 & -C_1S_2 & S_1 & l_1C_1 + l_2C_1C_2 \\ C_1S_1 & -S_1S_2 & -C_1 & l_1S_1 + l_2C_2S_1 \\ S_2 & C_2 & 0 & l_2S_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_2 = \begin{bmatrix} C_1C_2 & -C_1S_2 & S_1 \\ S_1C_2 & -S_1S_2 & -C_1 \end{bmatrix}$$

MATRIZ DE INÉRCIA

$$D = m[J_v]^T[J_v] + [J_w]^T[R][I][R]^T[J_w]$$

Para o 1 elo

$$D_1 = m_1[J_{v1}]^T[J_{v1}] + [J_{w1}]^T[R_1][I_1][R_1]^T[J_{w_1}]$$

$$D_1 = \begin{bmatrix} m_1 {l_1}^2 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} I_{yy1} & 0 \\ 0 & 0 \end{bmatrix}$$

$$D_1 = \begin{bmatrix} m_1 {l_1}^2 + I_{yy1} & 0 \\ 0 & 0 \end{bmatrix}$$

MATRIZ DE INÉRCIA

$$D = m[J_v]^T[J_v] + [J_w]^T[R][I][R]^T[J_w]$$

Para o 2 elo

$$D_2 = m_2[J_{v2}]^T[J_{v2}] + [J_{w2}]^T [R_2][I_2][R_2]^T[J_{w_2}]$$

$$D_2 = \begin{bmatrix} m_2(l_1 + l_2C_2)^2 & 0 \\ 0 & m_2{l_2}^2 \end{bmatrix} + \begin{bmatrix} I_{xx2} - {C_2}^2(I_{xx2} - I_{yy2}) & 0 \\ 0 & I_{zz2} \end{bmatrix}$$

$$D_2 = \begin{bmatrix} m_2(l_1 + l_2C_2)^2 + I_{xx2} - {C_2}^2(I_{xx2} - I_{yy2}) & 0 \\ 0 & m_2{l_2}^2 + I_{zz2} \end{bmatrix}$$

MATRIZ DE INÉRCIA

Somando os fatores D_1 e D_2

$$D = D_1 + D_2$$

$$D = \begin{bmatrix} I_{yy1} + m_1(l_1 + l_2C_2)^2 + m_2(l_1 + l_2C_2)^2 + I_{xx2} - {C_2}^2(I_{xx2} - I_{yy2}) & 0 \\ 0 & m_2{l_2}^2 + I_{zz2} \end{bmatrix}$$

$$D = \begin{bmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{bmatrix}$$

Matriz D utilizada para o calculo do Símbolo de Christoffel

$$d_{11} = I_{yy1} + m_1(l_1 + l_2C_2)^2 + m_2(l_1 + l_2C_2)^2 + I_{xx2} - {C_2}^2(I_{xx2} - I_{yy2})$$

$$d_{21} = d_{12} = 0$$

$$d_{22} = m_2{l_2}^2 + I_{zz2}$$

SÍMBOLO DE CHRISTOFFEL

$$c_{ijk} \equiv \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right) \quad C = \begin{bmatrix} C_{111} & C_{121} \\ C_{112} & C_{122} \end{bmatrix} \dot{\theta}_1 + \begin{bmatrix} C_{211} & C_{221} \\ C_{212} & C_{222} \end{bmatrix} \dot{\theta}_2$$

$$d_{11} = I_{yy1} + m_1(l_1 + l_2C_2)^2 + m_2(l_1 + l_2C_2)^2 + I_{xx2} - {C_2}^2(I_{xx2} - I_{yy2})$$

$$c_{111} = \frac{1}{2}\left\{\frac{\partial d11}{\partial \theta_1} + \frac{\partial d11}{\partial \theta_1} - \frac{\partial d11}{\partial \theta_1}\right\} = \frac{1}{2}\frac{\partial d11}{\partial \theta_1} = 0$$

$$c_{121} = \frac{1}{2}\left\{\frac{\partial d12}{\partial \theta_1} + \frac{\partial d11}{\partial \theta_2} - \frac{\partial d12}{\partial \theta_1}\right\} = \frac{1}{2}\frac{\partial d11}{\partial \theta_2}$$

$$c_{121} = \frac{I_{x2}sen(2\theta_2)}{2} - \frac{I_{y2}sen(2\theta_2)}{2} - \frac{l_2^2 m_2 sen(2\theta_2)}{2} - l_1 l_2 m_2 sen(\theta_2)$$

SÍMBOLO DE CHRISTOFFEL

$$d_{11} = I_{yy1} + m_1(l_1 + l_2C_2)^2 + m_2(l_1 + l_2C_2)^2 + I_{xx2} - {C_2}^2\left(I_{xx2} - I_{yy2}\right)$$

$$d_{21} = d_{12} = 0 \qquad\qquad d_{22} = m_2{l_2}^2 + I_{zz2}$$

$$c_{221} = \frac{1}{2}\left\{\frac{\partial d12}{\partial\theta_2} + \frac{\partial d12}{\partial\theta_2} - \frac{\partial d22}{\partial\theta_1}\right\} = 0$$

$$c_{112} = \frac{1}{2}\left\{\frac{\partial d21}{\partial\theta_1} + \frac{\partial d21}{\partial\theta_1} - \frac{\partial d11}{\partial\theta_2}\right\}$$

$$c_{112} = -\frac{I_{x2}sen(2\theta_2)}{2} + \frac{I_{y2}sen(2\theta_2)}{2} + \frac{l_2^2m_2sen(2\theta_2)}{2} + l_1l_2m_2sen(\theta_2)$$

$$\boxed{sen(2x) = 2sen(x)\cos(x)}$$

SÍMBOLO DE CHRISTOFFEL

$$d_{11} = I_{yy1} + m_1(l_1 + l_2C_2)^2 + m_2(l_1 + l_2C_2)^2 + I_{xx2} - {C_2}^2(I_{xx2} - I_{yy2})$$

$$d_{21} = d_{12} = 0 \qquad\qquad d_{22} = m_2{l_2}^2 + I_{zz2}$$

$$c_{122} = \frac{1}{2}\left\{\frac{\partial d22}{\partial \theta_1} + \frac{\partial d21}{\partial \theta_2} - \frac{\partial d12}{\partial \theta_2}\right\} = 0$$

$$c_{222} = \frac{1}{2}\left\{\frac{\partial d22}{\partial \theta_2} + \frac{\partial d22}{\partial \theta_2} - \frac{\partial d22}{\partial \theta_2}\right\} = 0$$

$$c_{212} = \frac{1}{2}\left\{\frac{\partial d21}{\partial \theta_2} + \frac{\partial d22}{\partial \theta_1} - \frac{\partial d21}{\partial \theta_2}\right\} = 0$$

$$c_{211} = \frac{1}{2}\left\{\frac{\partial d11}{\partial \theta_2} + \frac{\partial d12}{\partial \theta_1} - \frac{\partial d21}{\partial \theta_1}\right\} =$$

$$c_{211} = \frac{I_{x2}sen(2\theta_2)}{2} - \frac{I_{y2}sen(2\theta_2)}{2} - \frac{l_2^2 m_2 sen(2\theta_2)}{2} - l_1 l_2 m_2 sen(\theta_2)$$

SÍMBOLO DE CHRISTOFFEL

$$C = \begin{bmatrix} C_{111} & C_{121} \\ C_{112} & C_{122} \end{bmatrix} \dot{\theta}_1 + \begin{bmatrix} C_{211} & C_{221} \\ C_{212} & C_{222} \end{bmatrix} \dot{\theta}_2$$

$$C = \begin{bmatrix} 0 & C_{121} \\ C_{112} & 0 \end{bmatrix} \dot{\theta}_1 + \begin{bmatrix} C_{211} & 0 \\ 0 & 0 \end{bmatrix} \dot{\theta}_2$$

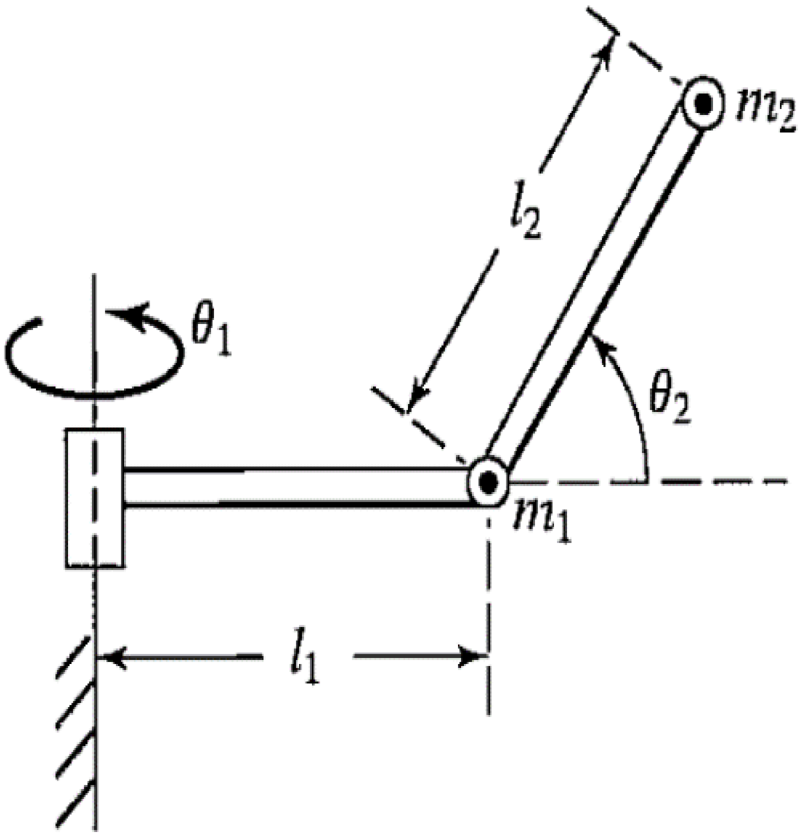
$$C = \begin{bmatrix} C_{211}\dot{\theta}_2 & C_{121}\dot{\theta}_1 \\ C_{112}\dot{\theta}_1 & 0 \end{bmatrix}$$

$$c_{121} = \frac{I_{x2}sen(2\theta_2)}{2} - \frac{I_{y2}sen(2\theta_2)}{2} - \frac{l_2^2 m_2 sen(2\theta_2)}{2} - l_1 l_2 m_2 sen(\theta_2)$$

$$c_{112} = -\frac{I_{x2}sen(2\theta_2)}{2} + \frac{I_{y2}sen(2\theta_2)}{2} + \frac{l_2^2 m_2 sen(2\theta_2)}{2} + l_1 l_2 m_2 sen(\theta_2)$$

$$c_{211} = \frac{I_{x2}sen(2\theta_2)}{2} - \frac{I_{y2}sen(2\theta_2)}{2} - \frac{l_2^2 m_2 sen(2\theta_2)}{2} - l_1 l_2 m_2 sen(\theta_2)$$

ENERGIA POTENCIAL



$$P = m_2 l_2 g sen(\theta_2)$$

$$g_1 = \frac{\partial P}{\partial \theta_1} = 0$$

$$g_2 = \frac{\partial P}{\partial \theta_2} = m_2 g l_2 \cos(\theta_2)$$

$$G = \begin{bmatrix} 0 \\ m_2 g l_2 \cos(\theta_2) \end{bmatrix}$$

EQUAÇÃO DE MOVIMENTO

$$D(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = \tau$$

$$\begin{bmatrix} I_{yy1} + \ m_1(l_1 + l_2C_2)^2 + m_2(l_1 + \ l_2C_2)^2 + I_{xx2} - {C_2}^2(I_{xx2} - I_{yy2}) & 0 \\ 0 & m_2{l_2}^2 + I_{zz2} \end{bmatrix} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix}$$

$$+ \begin{bmatrix} C_{211}\dot{\theta}_2 & C_{121}\dot{\theta}_1 \\ C_{112}\dot{\theta}_1 & 0 \end{bmatrix} \begin{Bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{Bmatrix} + \begin{bmatrix} 0 \\ m_2 g l_2 \cos(\theta_2) \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = \tau$$

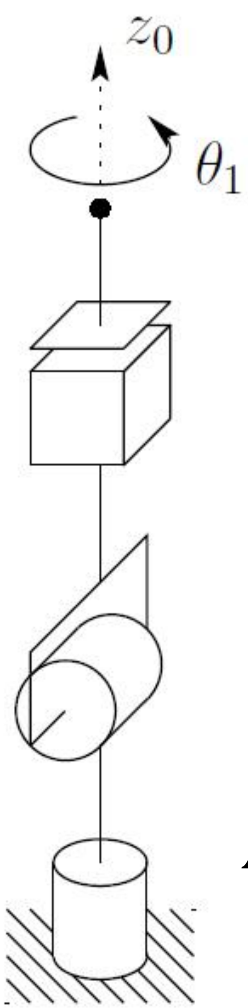
$$c_{121} = \frac{I_{x2} sen(2\theta_2)}{2} - \frac{I_{y2} sen(2\theta_2)}{2} - \frac{l_2^2 m_2 sen(2\theta_2)}{2} - \ l_1 l_2 m_2 sen(\theta_2)$$

$$c_{112} = -\frac{I_{x2} sen(2\theta_2)}{2} + \frac{I_{y2} sen(2\theta_2)}{2} + \frac{l_2^2 m_2 sen(2\theta_2)}{2} + \ l_1 l_2 m_2 sen(\theta_2)$$

$$c_{211} = \frac{I_{x2} sen(2\theta_2)}{2} - \frac{I_{y2} sen(2\theta_2)}{2} - \frac{l_2^2 m_2 sen(2\theta_2)}{2} - \ l_1 l_2 m_2 sen(\theta_2)$$

EXEMPLO 3

Os elos do manipulador possuem os tensores de inércia mostrados abaixo e possuem respectivamente as massas m_1, m_2e m_3. Os centros de massa dos braços estão localizados na extremidade das respectivas juntas



$$I_1 = \begin{bmatrix} I_{x1} & 0 & 0 \\ 0 & I_{y1} & 0 \\ 0 & 0 & I_{z1} \end{bmatrix}$$

$$I_2 = \begin{bmatrix} I_{x2} & 0 & 0 \\ 0 & I_{y2} & 0 \\ 0 & 0 & I_{z2} \end{bmatrix}$$

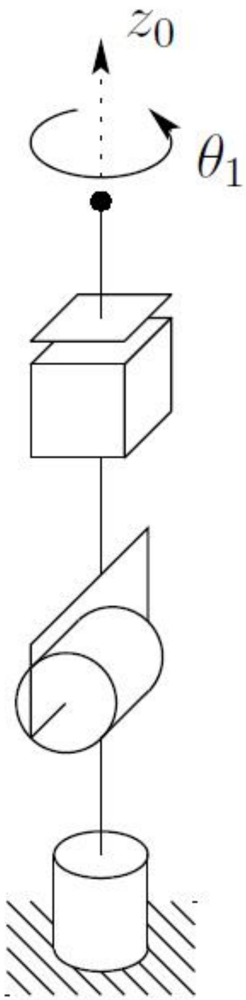
$$I_3 = \begin{bmatrix} I_{x3} & 0 & 0 \\ 0 & I_{y3} & 0 \\ 0 & 0 & I_{z3} \end{bmatrix}$$

Elo	a_i	α_i	d_i	θ_i
1	0	90	d_1	θ_1
2	0	-90	0	θ_2
3	0	0	d_2+d_3	0

$$A_1^0 = \begin{bmatrix} C_1 & 0 & S_1 & 0 \\ S_1 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2^1 = \begin{bmatrix} C_2 & 0 & -S_2 & 0 \\ S_2 & 0 & C_2 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_3^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_2 + d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

JACOBIANO



Origem

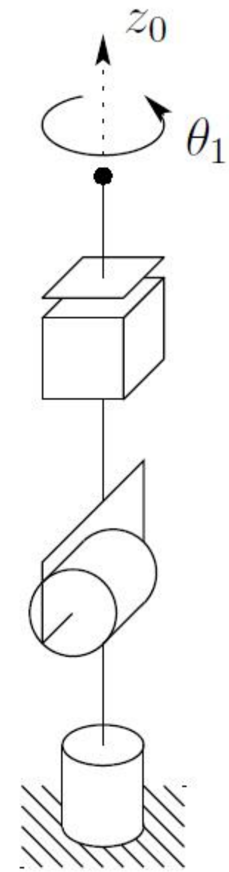
$$O_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad Z_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$T_1^0 = A_1^0 = \begin{array}{c} \begin{array}{cccc} & & \boldsymbol{Z_1} & \boldsymbol{o_1} \end{array} \\ \begin{bmatrix} C_1 & 0 & S_1 & 0 \\ S_1 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

$$T_2^0 = A_1^0 A_2^1 = \begin{array}{c} \begin{array}{cccc} & & \boldsymbol{Z_2} & \boldsymbol{o_2} \end{array} \\ \begin{bmatrix} C_1C_2 & -S_1 & -C_1S_2 & 0 \\ S_1C_2 & C_1 & -S_1S_2 & 0 \\ S_2 & 0 & C_2 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

$$T_3^0 = A_1^0 A_2^1 A_3^2 = \begin{array}{c} \begin{array}{cccc} & & & \boldsymbol{o_3} \end{array} \\ \begin{bmatrix} C_1C_2 & -S_1 & -C_1S_2 & -C_1S_2(d_2+d_3) \\ S_1C_2 & C_1 & -S_1S_2 & -S_1S_2(d_2+d_3) \\ S_2 & 0 & C_2 & d_1 + C_2(d_2+d_3) \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

JACOBIANO ELO 1



$$J_1 = \begin{bmatrix} Z_0 \times (O_1 - O_0) & 0 & 0 \\ Z_0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$J_{v1} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad J_{w1} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$A_1^0 = \begin{bmatrix} C_1 & 0 & S_1 & 0 \\ S_1 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad R_1 = \begin{bmatrix} C_1 & 0 & S_1 \\ S_1 & 0 & -C_1 \\ 0 & 1 & 0 \end{bmatrix}$$

MATRIZ DE INÉRCIA ELO 1

$$D = m[J_v]^T[J_v] + [J_w]^T[R][I][R]^T[J_w]$$

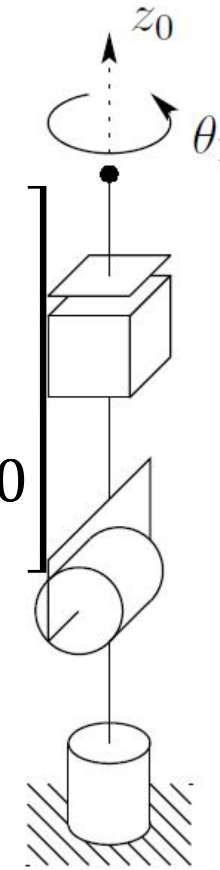
$$D_1 = m_1[J_{v1}]^T[J_{v1}] + [J_{w1}]^T[R_1][I_1][R_1]^T[J_{w_1}]$$

$$D_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} I_{y1} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$D_1 = \begin{bmatrix} I_{y1} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

JACOBIANO ELO 2

$$J_2 = \begin{bmatrix} Z_0 \times (O_2 - O_0) & Z_1 \times (O_2 - O_1) & 0 \\ Z_0 & Z_1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & S_1 & 0 \\ 0 & -C_1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$



$$J_{v2} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad J_{w2} = \begin{bmatrix} 0 & S_1 & 0 \\ 0 & -C_1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$A_2^0 = \begin{bmatrix} C_1C_2 & -S_1 & -C_1S_2 & 0 \\ S_1C_2 & C_1 & -S_1S_2 & 0 \\ S_2 & 0 & C_2 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad R_2 = \begin{bmatrix} C_1C_2 & -S_1 & -C_1S_2 \\ S_1C_2 & C_1 & -S_1S_2 \\ S_2 & 0 & C_2 \end{bmatrix}$$

MATRIZ DE INÉRCIA ELO 2

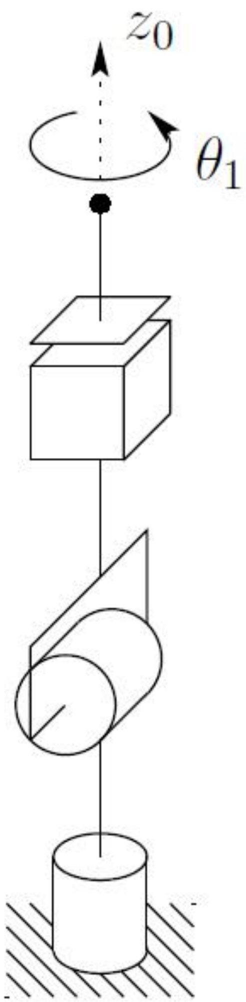
$$D = m[J_v]^T[J_v] + [J_w]^T[R][I][R]^T[J_w]$$

$$D_2 = m_2[J_{v2}]^T[J_{v2}] + [J_{w2}]^T\,[R_2][I_2][R_2]^T[J_{w_2}]$$

$$D_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} I_{z2} + I_{x2}S_2^2 - I_{z2}S_2^2 & 0 & 0 \\ 0 & I_{y2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$D_2 = \begin{bmatrix} I_{z2} + I_{x2}S_2^2 - I_{z2}S_2^2 & 0 & 0 \\ 0 & I_{y2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

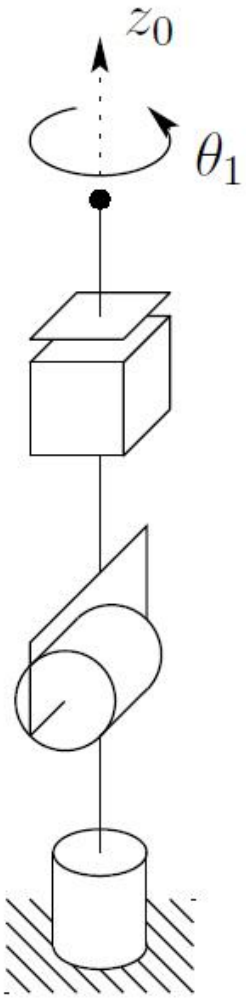
JACOBIANO ELO 3



$$J = \begin{bmatrix} Z_0 \times (O_3 - O_0) & Z_1 \times (O_3 - O_1) & Z_2 \\ Z_0 & Z_1 & [0\ 0\ 0]^T \end{bmatrix}$$

$$J = \begin{bmatrix} S_1S_2(d_2 + d_3) & -C_1C_2(d_2 + d_3) & -C_1S_2 \\ -C_1S_2(d_2 + d_3) & -C_2S_1(d_2 + d_3) & -S_1S_2 \\ 0 & -S_2(d_2 + d_3) & C_2 \\ 0 & S_1 & 0 \\ 0 & -C_1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

JACOBIANO ELO 3



$$J_{v3} = \begin{bmatrix} S_1S_2(d_2+d_3) & -C_1C_2(d_2+d_3) & -C_1S_2 \\ -C_1S_2(d_2+d_3) & -C_2S_1(d_2+d_3) & -S_1S_2 \\ 0 & -S_2(d_2+d_3) & C_2 \end{bmatrix}$$

$$J_{w3} = \begin{bmatrix} 0 & S_1 & 0 \\ 0 & -C_1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$A_3^0 = \begin{bmatrix} C_1C_2 & -S_1 & -C_1S_2 & -C_1S_2(d_2+d_3) \\ S_1C_2 & C_1 & -S_1S_2 & -S_1S_2(d_2+d_3) \\ S_2 & 0 & C_2 & d_1+C_2(d_2+d_3) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_3 = \begin{bmatrix} C_1C_2 & -S_1 & -C_1S_2 \\ S_1C_2 & C_1 & -S_1S_2 \\ S_2 & 0 & C_2 \end{bmatrix}$$

MATRIZ DE INÉRCIA ELO 3

$$D = m[J_v]^T[J_v] + [J_w]^T[R][I][R]^T[J_w]$$

$$D_3 = m_3[J_{v3}]^T[J_{v3}] + [J_{w3}]^T[R_3][I_3][R_3]^T[J_{w3}]$$

$$D_3 = \begin{bmatrix} m_3 S_2^2 (d_2 + d_3)^2 & 0 & 0 \\ 0 & m_3(d_2 + d_3)^2 & 0 \\ 0 & 0 & m_3 \end{bmatrix} + \begin{bmatrix} I_{z3} + I_{x3}S_2^2 - I_{z3}S_2^2 & 0 & 0 \\ 0 & I_{y3} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$D_3 = \begin{bmatrix} m_3 S_2^2 (d_2 + d_3)^2 + I_{z3} + I_{x3}S_2^2 - I_{z3}S_2^2 & 0 & 0 \\ 0 & m_3(d_2 + d_3)^2 + I_{y3} & 0 \\ 0 & 0 & m_3 \end{bmatrix}$$

MATRIZ DE INÉRCIA TOTAL

$$D = D_1 + D_2 + D_3$$

$$D = \begin{bmatrix} d_{11} & d_{12} & d_{13} \\ d_{21} & d_{22} & d_{23} \\ d_{31} & d_{32} & d_{33} \end{bmatrix}$$

$$d_{11} = I_{y1} + I_{z3} + I_{z2}C_2^2 + I_{x2}S_2^2 + I_{x3}S_2^2 - I_{z3}S_2^2 + m3S_2^2(d2 + d3)^2$$

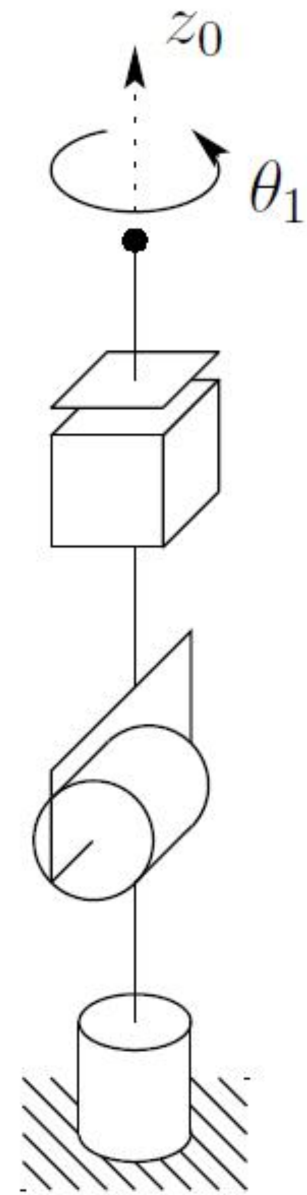
$$d_{22} = \mathrm{m_3 d_2^2} + \mathrm{2m_3 d_2 d_3} + \mathrm{m_3 d_3^2} + \mathrm{I}_{y2} + \mathrm{I}_{y3}$$

$$d_{33} = \mathrm{m_3}$$

$$d_{21} = d_{31} = d_{12} = d_{32} = d_{13} = d_{23} = 0$$

SÍMBOLO DE CHRISTOFFEL

$$c_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$



$$C = \begin{bmatrix} c_{111} & c_{121} & c_{131} \\ c_{112} & c_{122} & c_{132} \\ c_{113} & c_{123} & c_{133} \end{bmatrix} \dot{\theta}_1$$

$$+ \begin{bmatrix} c_{211} & c_{221} & c_{231} \\ c_{212} & c_{222} & c_{232} \\ c_{213} & c_{223} & c_{233} \end{bmatrix} \dot{\theta}_2$$

$$+ \begin{bmatrix} c_{311} & c_{321} & c_{331} \\ c_{312} & c_{322} & c_{332} \\ c_{313} & c_{323} & c_{333} \end{bmatrix} \dot{d}_3$$

SÍMBOLO DE CHRISTOFFEL JUNTA 1

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$C_1 = \begin{bmatrix} c_{111} & c_{121} & c_{131} \\ c_{112} & c_{122} & c_{132} \\ c_{113} & c_{123} & c_{133} \end{bmatrix} \dot{\theta}_1$$

$$c_{111} = \frac{1}{2}\left\{\frac{\partial d11}{\partial \theta_1} + \frac{\partial d11}{\partial \theta_1} - \frac{\partial d11}{\partial \theta_1}\right\} = 0$$

$$c_{112} = \frac{1}{2}\left\{\frac{\partial d21}{\partial \theta_1} + \frac{\partial d21}{\partial \theta_1} - \frac{\partial d11}{\partial \theta_2}\right\} =$$

$$c_{112} = \frac{-sen(2\theta_2)\,(m_3 d_2^2 + 2m_3 d_2 d_3 + m_3 d_3^2 \; + \; I_{x2} \; + \; I_{x3} \; - \; I_{z2} \; - \; I_{Z3})}{2}$$

$$\boxed{sen(2x) = 2sen(x)\cos(x)}$$

SÍMBOLO DE CHRISTOFFEL JUNTA 1

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right) \qquad C_1 = \begin{bmatrix} c_{111} & c_{121} & c_{131} \\ c_{112} & c_{122} & c_{132} \\ c_{113} & c_{123} & c_{133} \end{bmatrix} \dot{\theta}_1$$

$$c_{113} = \frac{1}{2}\left\{\frac{\partial d31}{\partial \theta_1} + \frac{\partial d31}{\partial \theta_1} - \frac{\partial d11}{\partial d_3}\right\} = -m_3 S_2^2 (d_2 + d_3)$$

$$c_{121} = \frac{1}{2}\left\{\frac{\partial d12}{\partial \theta_1} + \frac{\partial d11}{\partial \theta_2} - \frac{\partial d12}{\partial \theta_1}\right\}$$

$$c_{121} = \frac{sen(2\theta_2)\,(m_3 d_2^2 + 2m_3 d_2 d_3 + m_3 d_3^2 + I_{x2} + I_{x3} - I_{z2} - I_{Z3})}{2}$$

SÍMBOLO DE CHRISTOFFEL JUNTA 1

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right) \qquad C_1 = \begin{bmatrix} c_{111} & c_{121} & c_{131} \\ c_{112} & c_{122} & c_{132} \\ c_{113} & c_{123} & c_{133} \end{bmatrix} \dot{\theta}_1$$

$$c_{122} = \frac{1}{2}\left\{\frac{\partial d22}{\partial \theta_1} + \frac{\partial d21}{\partial \theta_2} - \frac{\partial d12}{\partial \theta_2}\right\} = 0$$

$$c_{123} = \frac{1}{2}\left\{\frac{\partial d32}{\partial \theta_1} + \frac{\partial d31}{\partial \theta_2} - \frac{\partial d12}{\partial d_3}\right\} = 0$$

$$c_{131} = \frac{1}{2}\left\{\frac{\partial d23}{\partial \theta_1} + \frac{\partial d11}{\partial d_3} - \frac{\partial d13}{\partial \theta_1}\right\} = m_3 S_2^2 (d_2 + d_3)$$

SÍMBOLO DE CHRISTOFFEL JUNTA 1

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right) \qquad C_1 = \begin{bmatrix} c_{111} & c_{121} & c_{131} \\ c_{112} & c_{122} & c_{132} \\ c_{113} & c_{123} & c_{133} \end{bmatrix} \dot{\theta}_1$$

$$c_{132} = \frac{1}{2}\left\{\frac{\partial d23}{\partial \theta_1} + \frac{\partial d21}{\partial d_3} - \frac{\partial d13}{\partial \theta_2}\right\} = 0$$

$$c_{133} = \frac{1}{2}\left\{\frac{\partial d33}{\partial \theta_1} + \frac{\partial d31}{\partial d_3} - \frac{\partial d13}{\partial d_3}\right\} = 0$$

SÍMBOLO DE CHRISTOFFEL JUNTA 2

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right) \qquad C_2 = \begin{bmatrix} c_{211} & c_{221} & c_{231} \\ c_{212} & c_{222} & c_{232} \\ c_{213} & c_{223} & c_{233} \end{bmatrix} \dot{\theta}_2$$

$$c_{211} = \frac{1}{2}\left\{\frac{\partial d11}{\partial \theta_2} + \frac{\partial d12}{\partial \theta_1} - \frac{\partial d21}{\partial \theta_1}\right\}$$

$$c_{211} = \frac{sen(2\theta_2)\ (m_3 d_2^2 + 2m_3 d_2 d_3 + m_3 d_3^2\ +\ I_{x2}\ +\ I_{x3}\ -\ I_{z2}\ -\ I_{Z3})}{2}$$

$$c_{212} = \frac{1}{2}\left\{\frac{\partial d21}{\partial \theta_2} + \frac{\partial d22}{\partial \theta_1} - \frac{\partial d21}{\partial \theta_2}\right\} = 0$$

$$c_{213} = \frac{1}{2}\left\{\frac{\partial d31}{\partial \theta_2} + \frac{\partial d32}{\partial \theta_1} - \frac{\partial d21}{\partial d_3}\right\} = 0$$

SÍMBOLO DE CHRISTOFFEL JUNTA 2

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right) \qquad C_2 = \begin{bmatrix} c_{211} & c_{221} & c_{231} \\ c_{212} & c_{222} & c_{232} \\ c_{213} & c_{223} & c_{233} \end{bmatrix} \dot{\theta}_2$$

$$c_{221} = \frac{1}{2}\left\{\frac{\partial d12}{\partial \theta_2} + \frac{\partial d12}{\partial \theta_2} - \frac{\partial d22}{\partial \theta_1}\right\} = 0$$

$$c_{222} = \frac{1}{2}\left\{\frac{\partial d22}{\partial \theta_2} + \frac{\partial d22}{\partial \theta_2} - \frac{\partial d22}{\partial \theta_2}\right\} = 0$$

$$c_{223} = \frac{1}{2}\left\{\frac{\partial d32}{\partial \theta_2} + \frac{\partial d32}{\partial \theta_2} - \frac{\partial d22}{\partial d_3}\right\} = -m_3(d_2 + d_3)$$

SÍMBOLO DE CHRISTOFFEL JUNTA 2

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right) \qquad C_2 = \begin{bmatrix} c_{211} & c_{221} & c_{231} \\ c_{212} & c_{222} & c_{232} \\ c_{213} & c_{223} & c_{233} \end{bmatrix} \dot{\theta}_2$$

$$c_{231} = \frac{1}{2}\left\{\frac{\partial d13}{\partial \theta_2} + \frac{\partial d12}{\partial d_3} - \frac{\partial d23}{\partial \theta_1}\right\} = 0$$

$$c_{232} = \frac{1}{2}\left\{\frac{\partial d23}{\partial \theta_2} + \frac{\partial d22}{\partial d_3} - \frac{\partial d23}{\partial \theta_2}\right\} = m_3(d_2 + d_3)$$

$$c_{233} = \frac{1}{2}\left\{\frac{\partial d33}{\partial \theta_2} + \frac{\partial d32}{\partial d_3} - \frac{\partial d23}{\partial d_3}\right\} = 0$$

SÍMBOLO DE CHRISTOFFEL JUNTA 3

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right) \qquad C_3 = \begin{bmatrix} c_{311} & c_{321} & c_{331} \\ c_{312} & c_{322} & c_{332} \\ c_{313} & c_{323} & c_{333} \end{bmatrix} \dot{d}_3$$

$$c_{311} = \frac{1}{2}\left\{\frac{\partial d11}{\partial d_3} + \frac{\partial d13}{\partial \theta_1} - \frac{\partial d31}{\partial \theta_1}\right\} = m_3 S_2^2 (d_2 + d_3)$$

$$c_{312} = \frac{1}{2}\left\{\frac{\partial d21}{\partial d_3} + \frac{\partial d23}{\partial \theta_1} - \frac{\partial d31}{\partial \theta_2}\right\} = 0$$

$$c_{313} = \frac{1}{2}\left\{\frac{\partial d31}{\partial d_3} + \frac{\partial d33}{\partial \theta_2} - \frac{\partial d31}{\partial d_3}\right\} = 0$$

SÍMBOLO DE CHRISTOFFEL JUNTA 3

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right) \qquad C_3 = \begin{bmatrix} c_{311} & c_{321} & c_{331} \\ c_{312} & c_{322} & c_{332} \\ c_{313} & c_{323} & c_{333} \end{bmatrix} \dot{d}_3$$

$$c_{321} = \frac{1}{2}\left\{\frac{\partial d12}{\partial d_3} + \frac{\partial d13}{\partial \theta_2} - \frac{\partial d32}{\partial \theta_1}\right\} = 0$$

$$c_{322} = \frac{1}{2}\left\{\frac{\partial d22}{\partial d_3} + \frac{\partial d23}{\partial \theta_2} - \frac{\partial d32}{\partial \theta_2}\right\} = m_3(d_2 + d_3)$$

$$c_{323} = \frac{1}{2}\left\{\frac{\partial d32}{\partial d_3} + \frac{\partial d33}{\partial \theta_2} - \frac{\partial d32}{\partial d_3}\right\} = 0$$

SÍMBOLO DE CHRISTOFFEL JUNTA 3

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right) \qquad C_3 = \begin{bmatrix} c_{311} & c_{321} & c_{331} \\ c_{312} & c_{322} & c_{332} \\ c_{313} & c_{323} & c_{333} \end{bmatrix} \dot{d}_3$$

$$c_{331} = \frac{1}{2}\left\{\frac{\partial d13}{\partial d_3} + \frac{\partial d13}{\partial d_3} - \frac{\partial d33}{\partial \theta_1}\right\} = 0$$

$$c_{332} = \frac{1}{2}\left\{\frac{\partial d23}{\partial d_3} + \frac{\partial d23}{\partial d_3} - \frac{\partial d33}{\partial \theta_2}\right\} = 0$$

$$c_{333} = \frac{1}{2}\left\{\frac{\partial d33}{\partial d_3} + \frac{\partial d33}{\partial d_3} - \frac{\partial d33}{\partial d_3}\right\} = 0$$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

Simplicações:

- Inverter os 2 primeiros dígitos gera resultados iguais

$C_{\alpha\beta\gamma} = C_{\beta\alpha\gamma}$ **Ex:** $C_{311} = C_{131}$

$C_{232} = C_{322}$

- Inverter os 2 últimos dígitos inverte o sinal do resultado

$C_{\alpha\beta\gamma} = -C_{\alpha\gamma\beta}$ **Ex:** $C_{223} = -C_{232}$

$C_{112} = -C_{121}$

SÍMBOLO DE CHRISTOFFEL

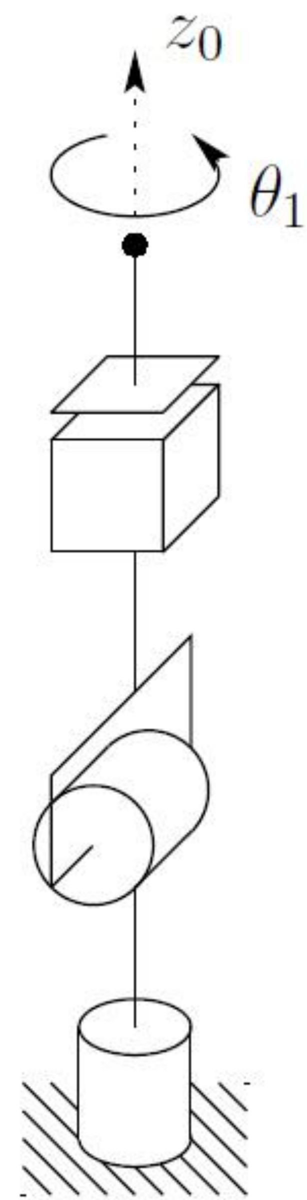
$$C = \begin{bmatrix} c_{111} & c_{121} & c_{131} \\ c_{112} & c_{122} & c_{132} \\ c_{113} & c_{123} & c_{133} \end{bmatrix} \dot{\theta}_1 + \begin{bmatrix} c_{211} & c_{221} & c_{231} \\ c_{212} & c_{222} & c_{232} \\ c_{213} & c_{223} & c_{233} \end{bmatrix} \dot{\theta}_2 + \begin{bmatrix} c_{311} & c_{321} & c_{331} \\ c_{312} & c_{322} & c_{332} \\ c_{313} & c_{323} & c_{333} \end{bmatrix} \dot{d}_3$$

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$$

ENERGIA POTENCIAL

$$P_1 = m_1 g d_1$$
$$P_2 = m_2 g(d_1 + C_2 d_2)$$
$$P_3 = m_3 g(d_1 + C_2(d_2 + d_3))$$

$$P = P_1 + P_2 + P_3$$

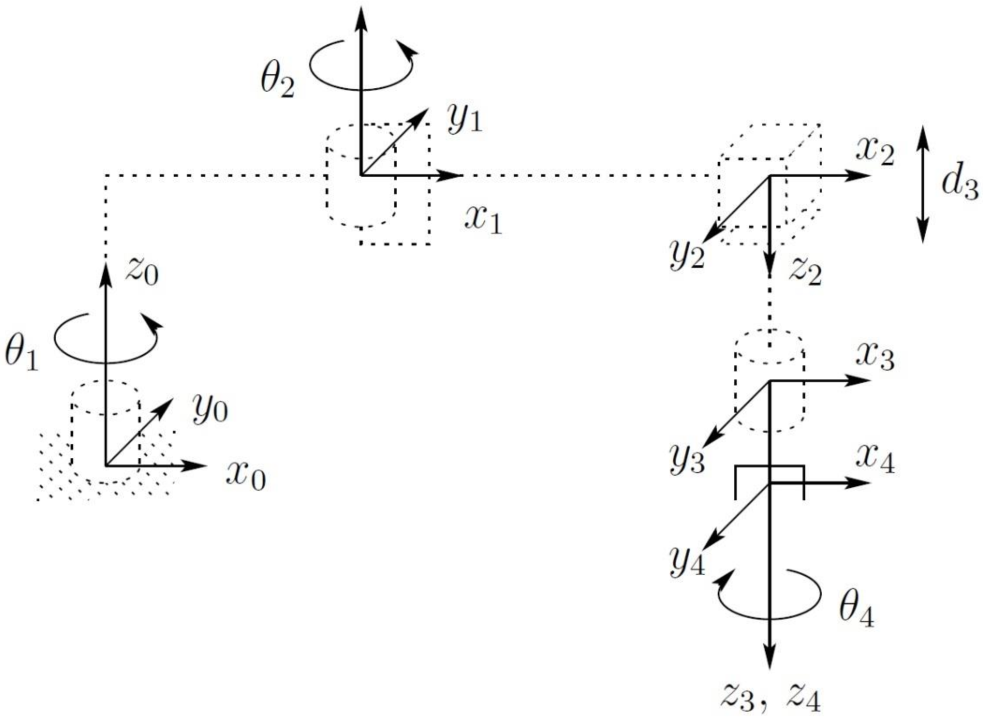


$$g_1 = \frac{\partial P}{\partial \theta_1} = 0 \qquad g_3 = \frac{\partial P}{\partial d_3} = g m_3 C_2$$

$$g_2 = \frac{\partial P}{\partial \theta_2} = -g S_2(d_2 m_2 + d_2 m_3 + d_3 m_3)$$

$$g = \begin{bmatrix} g_1 \\ g_2 \\ g_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -g S_2(d_2 m_2 + d_2 m_3 + d_3 m_3) \\ g m_3 C_2 \end{bmatrix}$$

EXEMPLO MANIPULADOR SCARA



$$I_1 = \begin{bmatrix} I_{xx1} & 0 & 0 \\ 0 & I_{yy1} & 0 \\ 0 & 0 & I_{zz1} \end{bmatrix} \quad I_2 = \begin{bmatrix} I_{xx2} & 0 & 0 \\ 0 & I_{yy2} & 0 \\ 0 & 0 & I_{zz2} \end{bmatrix}$$

$$I_3 = \begin{bmatrix} I_{xx3} & 0 & 0 \\ 0 & I_{yy3} & 0 \\ 0 & 0 & I_{zz3} \end{bmatrix} \quad I_4 = \begin{bmatrix} I_{xx4} & 0 & 0 \\ 0 & I_{yy4} & 0 \\ 0 & 0 & I_{zz4} \end{bmatrix}$$

Elo	a_i	α_i	d_i	θ_i
1	a_1	0	d_1	θ_1
2	a_2	180	0	θ_2
3	0	0	d_3	0
4	0	0	d_4	θ_4

EXEMPLO MANIPULADOR SCARA

$$A_1 = \begin{bmatrix} c_1 & -s_1 & 0 & a_1c_1 \\ s_1 & c_1 & 0 & a_1s_1 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, A_2 = \begin{bmatrix} c_2 & s_2 & 0 & a_2c_2 \\ s_2 & -c_2 & 0 & a_2s_2 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, A_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}, A_4 = \begin{bmatrix} c_4 & -s_4 & 0 & 0 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 1 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_1^0 = \begin{bmatrix} C_1 & -S_1 & 0 & a_1C_1 \\ S_1 & C_1 & 0 & a_1S_1 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad T_2^0 = \begin{bmatrix} C_{12} & S_{12} & 0 & a_1C_1 + a_2C_{12} \\ S_{12} & -C_{12} & 0 & a_1S_1 + a_2S_{12} \\ 0 & 0 & -1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_3^0 = \begin{bmatrix} C_{12} & S_{12} & 0 & a_1C_1 + a_2C_{12} \\ S_{12} & -C_{12} & 0 & a_1S_1 + a_2S_{12} \\ 0 & 0 & -1 & d_1 - d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_4^0 = \begin{bmatrix} C_{12}C_4 + S_{12}S_4 & -C_{12}S_4 + S_{12}C_4 & 0 & a_1C_1 + a_2C_{12} \\ S_{12}C_4 - C_{12}S_4 & -S_{12}S_4 - C_{12}C_4 & 0 & a_1S_1 + a_2S_{12} \\ 0 & 0 & -1 & d_1 - d_3 - d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

EQUAÇÃO DE MOVIMENTO

$$D(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = \tau$$

Matriz de Inércia $$D = m[J_v]^T[J_v] + [J_w]^T[R][I][R]^T[J_w]$$

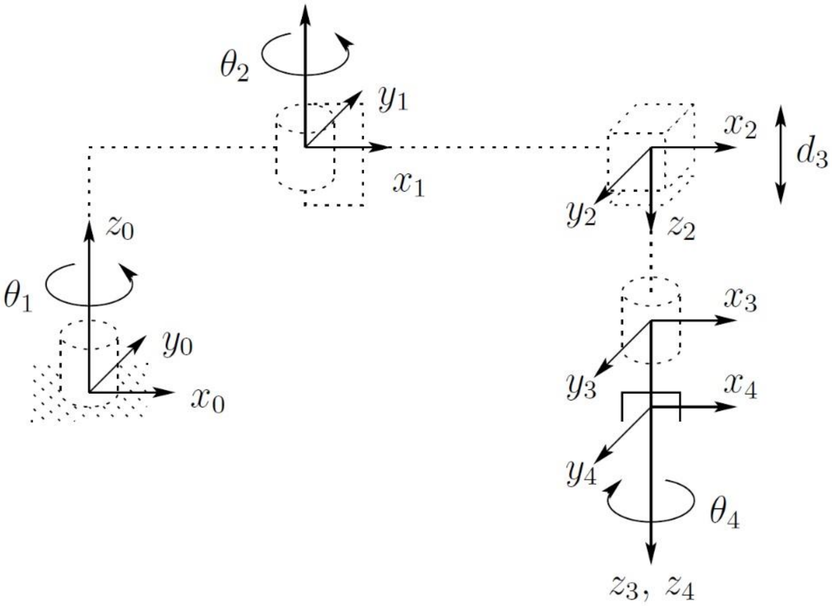
$$J_4 = \begin{bmatrix} Z_0 \times (O_4 - O_0) & Z_1 \times (O_4 - O_1) & Z_2 & Z_3 \times (O_4 - O_3) \\ Z_0 & Z_1 & 0 & Z_3 \end{bmatrix}$$

$$J_3 = \begin{bmatrix} Z_0 \times (O_3 - O_0) & Z_1 \times (O_3 - O_1) & Z_2 & 0 \\ Z_0 & Z_1 & 0 & 0 \end{bmatrix}$$

$$J_2 = \begin{bmatrix} Z_0 \times (O_2 - O_0) & Z_1 \times (O_2 - O_1) & 0 & 0 \\ Z_0 & Z_1 & 0 & 0 \end{bmatrix}$$

$$J_1 = \begin{bmatrix} Z_0 \times (O_1 - O_0) & 0 & 0 & 0 \\ Z_0 & 0 & 0 & 0 \end{bmatrix}$$

JACOBIANO ELO 1



$$J_1 = \begin{bmatrix} Z_0 \times (O_1 - O_0) & 0 & 0 & 0 \\ Z_0 & 0 & 0 & 0 \end{bmatrix}$$

$$J_{v1} = \begin{bmatrix} -a_1 S_1 & 0 & 0 & 0 \\ a_1 C_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad J_{w1} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$A_1^0 = \begin{bmatrix} C_1 & -S_1 & 0 & a_1 C_1 \\ S_1 & C_1 & 0 & a_1 S_1 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad R_1 = \begin{bmatrix} C_1 & -S_1 & 0 \\ S_1 & C_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

MATRIZ DE INÉRCIA ELO 1

$$D = m[J_v]^T[J_v] + [J_w]^T[R][I][R]^T[J_w]$$

$$D_1 = m_1[J_{v1}]^T[J_{v1}] + [J_{w1}]^T\,[R_1][I_1][R_1]^T[J_{w_1}]$$

$$D_1 = \begin{bmatrix} a_1^2 m_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} I_{Z1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$D_1 = \begin{bmatrix} a_1^2 m_1 + I_{Z1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

JACOBIANO ELO 2

$$J_2 = \begin{bmatrix} Z_0 \times (O_2 - O_0) & Z_1 \times (O_2 - O_1) & 0 & 0 \\ Z_0 & Z_1 & 0 & 0 \end{bmatrix}$$

$$J_{v2} = \begin{bmatrix} -a_2 S_{12} - a_1 S_1 & -a_2 S_{12} & 0 & 0 \\ a_2 C_{12} + a_1 C_1 & a_2 C_{12} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$J_{w2} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

$$A_2^0 = \begin{bmatrix} C_{12} & S_{12} & 0 & a_1 C_1 + a_2 C_{12} \\ S_{12} & -C_{12} & 0 & a_1 S_1 + a_2 S_{12} \\ 0 & 0 & -1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad R_2 = \begin{bmatrix} C_{12} & S_{12} & 0 \\ S_{12} & -C_{12} & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

MATRIZ DE INÉRCIA ELO 2

$$D_2 = m_2[J_{v2}]^T[J_{v2}] + [J_{w2}]^T\,[R_2][I_2][R_2]^T[\,J_{w_2}]$$

$$D_2 = \begin{bmatrix} m_2(a_1^2 + 2C_2a_1a_2 + a_2^2) & a_2m_2(a_2 + a_1C_2) & 0 & 0 \\ a_2m_2(a_2 + a_1C_2) & a_2^2m_2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} I_{Z2} & I_{Z2} & 0 & 0 \\ I_{Z2} & I_{Z2} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$D_2 = \begin{bmatrix} m_2(a_1^2 + 2C_2a_1a_2 + a_2^2) + I_{Z2} & a_2m_2(a_2 + a_1C_2) + I_{Z2} & 0 & 0 \\ a_2m_2(a_2 + a_1C_2) + I_{Z2} & a_2^2m_2 + I_{Z2} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

JACOBIANO ELO 3

$$J_3 = \begin{bmatrix} Z_0 \times (O_3 - O_0) & Z_1 \times (O_3 - O_1) & Z_2 & 0 \\ Z_0 & Z_1 & 0 & 0 \end{bmatrix}$$

$$J_{v3} = \begin{bmatrix} -a_2S_{12} - a_1S_1 & -a_2S_{12} & 0 & 0 \\ a_2C_{12} + a_1C_1 & a_2C_{12} & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

$$J_{w3} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

$$A_3^0 = \begin{bmatrix} C_{12} & S_{12} & 0 & a_1C_1 + a_2C_{12} \\ S_{12} & -C_{12} & 0 & a_1S_1 + a_2S_{12} \\ 0 & 0 & -1 & d_1 - d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad R_3 = \begin{bmatrix} C_{12} & S_{12} & 0 \\ S_{12} & -C_{12} & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

MATRIZ DE INÉRCIA ELO 3

$$D_3 = m_3[J_{v3}]^T[J_{v3}] + [J_{w3}]^T\,[R_3][I_3][R_3]^T[\,J_{w3}]$$

$$D_3 = \begin{bmatrix} m_3(a_1^2 + 2C_2a_1a_2 + a_2^2) & a_2m_3(a_2 + a_1C_2) & 0 & 0 \\ a_2m_3(a_2 + a_1C_2) & a_2^2m_3 & 0 & 0 \\ 0 & 0 & m_3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} I_{Z3} & I_{Z3} & 0 & 0 \\ I_{Z3} & I_{Z3} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$D_3 = \begin{bmatrix} m_3(a_1^2 + 2C_2a_1a_2 + a_2^2) + I_{Z3} & a_2m_3(a_2 + a_1C_2) + I_{Z3} & 0 & 0 \\ a_2m_3(a_2 + a_1C_2) + I_{Z3} & a_2^2m_3 + I_{Z3} & 0 & 0 \\ 0 & 0 & m_3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

JACOBIANO ELO 4

$$J_4 = \begin{bmatrix} Z_0 \times (O_4 - O_0) & Z_1 \times (O_4 - O_1) & Z_2 & Z_3 \times (O_4 - O_3) \\ Z_0 & Z_1 & 0 & Z_3 \end{bmatrix}$$

$$J_{v4} = \begin{bmatrix} -a_2 S_{12} - a_1 S_1 & -a_2 S_{12} & 0 & 0 \\ a_2 C_{12} + a_1 C_1 & a_2 C_{12} & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix} \qquad J_{w4} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -1 \end{bmatrix}$$

$$A_4^0 = \begin{bmatrix} C_{12}C_4 + S_{12}S_4 & -C_{12}S_4 + S_{12}C_4 & 0 & a_1 C_1 + a_2 C_{12} \\ S_{12}C_4 - C_{12}S_4 & -S_{12}S_4 - C_{12}C_4 & 0 & a_1 S_1 + a_2 S_{12} \\ 0 & 0 & -1 & d_1 - d_3 - d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_4 = \begin{bmatrix} C_{12}C_4 + S_{12}S_4 & -C_{12}S_4 + S_{12}C_4 & 0 \\ S_{12}C_4 - C_{12}S_4 & -S_{12}S_4 - C_{12}C_4 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

MATRIZ DE INÉRCIA ELO 4

$$D_4 = m_4[J_{v4}]^T[J_{v4}] + [J_{w4}]^T\,[R_4][I_4][R_4]^T[\,J_{w_4}]$$

$$D_4 = \begin{bmatrix} m_4(a_1^2 + 2C_2a_1a_2 + a_2^2) & a_2m_4(a_2 + a_1C_2) & 0 & 0 \\ a_2m_4(a_2 + a_1C_2) & a_2^2m_4 & 0 & 0 \\ 0 & 0 & m_4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} I_{Z4} & I_{Z4} & 0 & -I_{Z4} \\ I_{Z4} & I_{Z4} & 0 & -I_{Z4} \\ 0 & 0 & 0 & 0 \\ -I_{Z4} & -I_{Z4} & 0 & -I_{Z4} \end{bmatrix}$$

$$D_4 = \begin{bmatrix} m_4(a_1^2 + 2C_2a_1a_2 + a_2^2) + I_{Z4} & a_2m_4(a_2 + a_1C_2) + I_{Z4} & 0 & -I_{Z4} \\ a_2m_4(a_2 + a_1C_2) + I_{Z4} & a_2^2m_4 + I_{Z4} & 0 & -I_{Z4} \\ 0 & 0 & m_4 & 0 \\ -I_{Z4} & -I_{Z4} & 0 & -I_{Z4} \end{bmatrix}$$

MATRIZ DE INÉRCIA TOTAL

$$D = D_1 + D_2 + D_3 + D_4$$

$$D = \begin{bmatrix} d_{11} & d_{12} & d_{13} & d_{14} \\ d_{21} & d_{22} & d_{23} & d_{24} \\ d_{31} & d_{32} & d_{33} & d_{34} \\ d_{41} & d_{42} & d_{43} & d_{44} \end{bmatrix}$$

$$d_{11} = I_{Z1} + I_{Z2} + I_{Z3} + I_{Z4} + a_1^2(m_1 + m_2 + m_3 + m_4) + a_2^2(m_2 + m_3 + m_4) + 2a_1a_2C_2(m_2 + m_3 + m_4)$$

$$d_{12} = I_{Z2} + I_{Z3} + I_{Z4} + a_2^2(m_2 + m_3 + m_4) + a_1a_2C_2(m_2 + m_3 + m_4)$$

MATRIZ DE INÉRCIA TOTAL

$$d_{13} = 0 \qquad\qquad d_{14} = -I_{Z4}$$

$$d_{21} = I_{Z2} + I_{Z3} + I_{Z4} + a_2^2(m_2 + m_3 + m_4) + a_1 a_2 C_2 (m_2 + m_3 + m_4)$$

$$d_{22} = I_{Z2} + I_{Z3} + I_{Z4} + a_2^2(m_2 + m_3 + m_4)$$

$$d_{23} = 0 \qquad\qquad d_{24} = -I_{Z4}$$

$$d_{31} = 0 \qquad d_{32} = 0 \qquad d_{33} = m_3 + m_4 \qquad d_{34} = 0$$

$$d_{41} = -I_{Z4} \qquad d_{42} = -I_{Z4} \qquad d_{43} = 0 \qquad d_{44} = I_{Z4}$$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$C = \begin{bmatrix} c_{111} & c_{121} & c_{131} & c_{141} \\ c_{112} & c_{122} & c_{132} & c_{142} \\ c_{113} & c_{123} & c_{133} & c_{143} \\ c_{114} & c_{124} & c_{134} & c_{144} \end{bmatrix} \dot{\theta}_1 + \begin{bmatrix} c_{211} & c_{221} & c_{231} & c_{241} \\ c_{212} & c_{222} & c_{232} & c_{242} \\ c_{213} & c_{223} & c_{233} & c_{243} \\ c_{214} & c_{224} & c_{234} & c_{244} \end{bmatrix} \dot{\theta}_2$$

$$+ \begin{bmatrix} c_{311} & c_{321} & c_{331} & c_{341} \\ c_{312} & c_{322} & c_{332} & c_{342} \\ c_{313} & c_{323} & c_{333} & c_{343} \\ c_{314} & c_{324} & c_{334} & c_{344} \end{bmatrix} \dot{d}_3 + \begin{bmatrix} c_{411} & c_{421} & c_{431} & c_{441} \\ c_{412} & c_{422} & c_{432} & c_{442} \\ c_{413} & c_{423} & c_{433} & c_{443} \\ c_{414} & c_{424} & c_{434} & c_{444} \end{bmatrix} \dot{\theta}_4$$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$c_{111} = \frac{1}{2}\left\{\frac{\partial d11}{\partial \theta_1} + \frac{\partial d11}{\partial \theta_1} - \frac{\partial d11}{\partial \theta_1}\right\} = 0$$

$$\boxed{\begin{aligned} C_{\alpha\beta\gamma} &= C_{\beta\alpha\gamma} \\ C_{\alpha\beta\gamma} &= -C_{\alpha\gamma\beta} \end{aligned}}$$

$$c_{112} = \frac{1}{2}\left\{\frac{\partial d21}{\partial \theta_1} + \frac{\partial d21}{\partial \theta_1} - \frac{\partial d11}{\partial \theta_2}\right\} = a_1 a_2 S_2 (m_2 + m_3 + m_4)$$

$$c_{121} = -c_{112} = -a_1 a_2 S_2 (m_2 + m_3 + m_4)$$

$$c_{211} = c_{121} = -a_1 a_2 S_2 (m_2 + m_3 + m_4)$$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$C = \begin{bmatrix} c_{111} & c_{121} & c_{131} & c_{141} \\ c_{112} & c_{122} & c_{132} & c_{142} \\ c_{113} & c_{123} & c_{133} & c_{143} \\ c_{114} & c_{124} & c_{134} & c_{144} \end{bmatrix} \dot{\theta}_1 + \begin{bmatrix} c_{211} & c_{221} & c_{231} & c_{241} \\ c_{212} & c_{222} & c_{232} & c_{242} \\ c_{213} & c_{223} & c_{233} & c_{243} \\ c_{214} & c_{224} & c_{234} & c_{244} \end{bmatrix} \dot{\theta}_2$$

$$+ \begin{bmatrix} c_{311} & c_{321} & c_{331} & c_{341} \\ c_{312} & c_{322} & c_{332} & c_{342} \\ c_{313} & c_{323} & c_{333} & c_{343} \\ c_{314} & c_{324} & c_{334} & c_{344} \end{bmatrix} \dot{d}_3 + \begin{bmatrix} c_{411} & c_{421} & c_{431} & c_{441} \\ c_{412} & c_{422} & c_{432} & c_{442} \\ c_{413} & c_{423} & c_{433} & c_{443} \\ c_{414} & c_{424} & c_{434} & c_{444} \end{bmatrix} \dot{\theta}_4$$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$\boxed{\begin{aligned} C_{\alpha\beta\gamma} &= C_{\beta\alpha\gamma} \\ C_{\alpha\beta\gamma} &= -C_{\alpha\gamma\beta} \end{aligned}}$$

$$c_{113} = \frac{1}{2}\left\{\frac{\partial d31}{\partial \theta_1} + \frac{\partial d31}{\partial \theta_1} - \frac{\partial d11}{\partial d_3}\right\} = 0$$

$$c_{131} = -c_{113} = 0 \qquad c_{311} = c_{131} = 0$$

$$c_{114} = \frac{1}{2}\left\{\frac{\partial d41}{\partial \theta_1} + \frac{\partial d41}{\partial \theta_1} - \frac{\partial d11}{\partial \theta_4}\right\} = 0$$

$$c_{141} = -c_{114} = 0 \qquad c_{411} = c_{141} = 0$$

$$c_{122} = \frac{1}{2}\left\{\frac{\partial d22}{\partial \theta_1} + \frac{\partial d21}{\partial \theta_2} - \frac{\partial d12}{\partial \theta_2}\right\} = 0$$

$$c_{212} = c_{122} = 0 \qquad c_{221} = -c_{212} = 0$$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$C = \begin{bmatrix} c_{111} & c_{121} & c_{131} & c_{141} \\ c_{112} & c_{122} & c_{132} & c_{142} \\ c_{113} & c_{123} & c_{133} & c_{143} \\ c_{114} & c_{124} & c_{134} & c_{144} \end{bmatrix} \dot{\theta}_1 + \begin{bmatrix} c_{211} & c_{221} & c_{231} & c_{241} \\ c_{212} & c_{222} & c_{232} & c_{242} \\ c_{213} & c_{223} & c_{233} & c_{243} \\ c_{214} & c_{224} & c_{234} & c_{244} \end{bmatrix} \dot{\theta}_2$$

$$+ \begin{bmatrix} c_{311} & c_{321} & c_{331} & c_{341} \\ c_{312} & c_{322} & c_{332} & c_{342} \\ c_{313} & c_{323} & c_{333} & c_{343} \\ c_{314} & c_{324} & c_{334} & c_{344} \end{bmatrix} \dot{d}_3 + \begin{bmatrix} c_{411} & c_{421} & c_{431} & c_{441} \\ c_{412} & c_{422} & c_{432} & c_{442} \\ c_{413} & c_{423} & c_{433} & c_{443} \\ c_{414} & c_{424} & c_{434} & c_{444} \end{bmatrix} \dot{\theta}_4$$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$c_{123} = \frac{1}{2}\left\{\frac{\partial d32}{\partial \theta_1} + \frac{\partial d31}{\partial \theta_2} - \frac{\partial d12}{\partial d_3}\right\} = 0$$

$$\boxed{\begin{aligned} C_{\alpha\beta\gamma} &= C_{\beta\alpha\gamma} \\ C_{\alpha\beta\gamma} &= -C_{\alpha\gamma\beta} \end{aligned}}$$

$c_{132} = -c_{123} = 0$ $\quad$ $c_{213} = c_{123} = 0$

$c_{231} = -c_{213} = 0$ $\quad$ $c_{321} = c_{231} = 0$ $\quad$ $c_{312} = -c_{321} = 0$

$$c_{124} = \frac{1}{2}\left\{\frac{\partial d42}{\partial \theta_1} + \frac{\partial d41}{\partial \theta_2} - \frac{\partial d12}{\partial \theta_4}\right\} = 0$$

$c_{142} = -c_{124} = 0$ $\quad$ $c_{214} = c_{124} = 0$

$c_{241} = -c_{214} = 0$ $\quad$ $c_{421} = c_{241} = 0$ $\quad$ $c_{412} = -c_{421} = 0$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$C = \begin{bmatrix} c_{111} & c_{121} & c_{131} & c_{141} \\ c_{112} & c_{122} & c_{132} & c_{142} \\ c_{113} & c_{123} & c_{133} & c_{143} \\ c_{114} & c_{124} & c_{134} & c_{144} \end{bmatrix} \dot{\theta}_1 + \begin{bmatrix} c_{211} & c_{221} & c_{231} & c_{241} \\ c_{212} & c_{222} & c_{232} & c_{242} \\ c_{213} & c_{223} & c_{233} & c_{243} \\ c_{214} & c_{224} & c_{234} & c_{244} \end{bmatrix} \dot{\theta}_2$$

$$+ \begin{bmatrix} c_{311} & c_{321} & c_{331} & c_{341} \\ c_{312} & c_{322} & c_{332} & c_{342} \\ c_{313} & c_{323} & c_{333} & c_{343} \\ c_{314} & c_{324} & c_{334} & c_{344} \end{bmatrix} \dot{d}_3 + \begin{bmatrix} c_{411} & c_{421} & c_{431} & c_{441} \\ c_{412} & c_{422} & c_{432} & c_{442} \\ c_{413} & c_{423} & c_{433} & c_{443} \\ c_{414} & c_{424} & c_{434} & c_{444} \end{bmatrix} \dot{\theta}_4$$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$c_{133} = \frac{1}{2}\left\{\frac{\partial d33}{\partial \theta_1} + \frac{\partial d31}{\partial d_3} - \frac{\partial d13}{\partial d_3}\right\} = 0$$

$$\boxed{\begin{aligned} C_{\alpha\beta\gamma} &= C_{\beta\alpha\gamma} \\ C_{\alpha\beta\gamma} &= -C_{\alpha\gamma\beta} \end{aligned}}$$

$c_{313} = c_{133} = 0$ $\qquad$ $c_{331} = -c_{313} = 0$

$$c_{134} = \frac{1}{2}\left\{\frac{\partial d43}{\partial \theta_1} + \frac{\partial d41}{\partial d_3} - \frac{\partial d13}{\partial \theta_4}\right\} = 0$$

$c_{143} = -c_{134} = 0$ $\qquad$ $c_{314} = c_{134} = 0$

$c_{341} = -c_{314} = 0$ $\qquad$ $c_{431} = c_{341} = 0$ $\qquad$ $c_{413} = -c_{431} = 0$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$C = \begin{bmatrix} c_{111} & c_{121} & c_{131} & c_{141} \\ c_{112} & c_{122} & c_{132} & c_{142} \\ c_{113} & c_{123} & c_{133} & c_{143} \\ c_{114} & c_{124} & c_{134} & c_{144} \end{bmatrix}\dot{\theta}_1 + \begin{bmatrix} c_{211} & c_{221} & c_{231} & c_{241} \\ c_{212} & c_{222} & c_{232} & c_{242} \\ c_{213} & c_{223} & c_{233} & c_{243} \\ c_{214} & c_{224} & c_{234} & c_{244} \end{bmatrix}\dot{\theta}_2$$

$$+ \begin{bmatrix} c_{311} & c_{321} & c_{331} & c_{341} \\ c_{312} & c_{322} & c_{332} & c_{342} \\ c_{313} & c_{323} & c_{333} & c_{343} \\ c_{314} & c_{324} & c_{334} & c_{344} \end{bmatrix}\dot{d}_3 + \begin{bmatrix} c_{411} & c_{421} & c_{431} & c_{441} \\ c_{412} & c_{422} & c_{432} & c_{442} \\ c_{413} & c_{423} & c_{433} & c_{443} \\ c_{414} & c_{424} & c_{434} & c_{444} \end{bmatrix}\dot{\theta}_4$$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$\boxed{\begin{aligned} C_{\alpha\beta\gamma} &= C_{\beta\alpha\gamma} \\ C_{\alpha\beta\gamma} &= -C_{\alpha\gamma\beta} \end{aligned}}$$

$$c_{144} = \frac{1}{2}\left\{\frac{\partial d44}{\partial \theta_1} + \frac{\partial d41}{\partial \theta_4} - \frac{\partial d14}{\partial \theta_4}\right\} = 0$$

$$c_{414} = c_{144} = 0 \qquad c_{441} = -c_{414} = 0$$

$$c_{222} = \frac{1}{2}\left\{\frac{\partial d22}{\partial \theta_2} + \frac{\partial d22}{\partial \theta_2} - \frac{\partial d22}{\partial \theta_2}\right\} = 0$$

$$c_{223} = \frac{1}{2}\left\{\frac{\partial d32}{\partial \theta_2} + \frac{\partial d32}{\partial \theta_2} - \frac{\partial d22}{\partial d_3}\right\} = 0$$

$$c_{232} = -c_{223} = 0 \qquad c_{322} = c_{232} = 0$$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$C = \begin{bmatrix} c_{111} & c_{121} & c_{131} & c_{141} \\ c_{112} & c_{122} & c_{132} & c_{142} \\ c_{113} & c_{123} & c_{133} & c_{143} \\ c_{114} & c_{124} & c_{134} & c_{144} \end{bmatrix} \dot{\theta}_1 + \begin{bmatrix} c_{211} & c_{221} & c_{231} & c_{241} \\ c_{212} & c_{222} & c_{232} & c_{242} \\ c_{213} & c_{223} & c_{233} & c_{243} \\ c_{214} & c_{224} & c_{234} & c_{244} \end{bmatrix} \dot{\theta}_2$$

$$+ \begin{bmatrix} c_{311} & c_{321} & c_{331} & c_{341} \\ c_{312} & c_{322} & c_{332} & c_{342} \\ c_{313} & c_{323} & c_{333} & c_{343} \\ c_{314} & c_{324} & c_{334} & c_{344} \end{bmatrix} \dot{d}_3 + \begin{bmatrix} c_{411} & c_{421} & c_{431} & c_{441} \\ c_{412} & c_{422} & c_{432} & c_{442} \\ c_{413} & c_{423} & c_{433} & c_{443} \\ c_{414} & c_{424} & c_{434} & c_{444} \end{bmatrix} \dot{\theta}_4$$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$c_{224} = \frac{1}{2}\left\{\frac{\partial d42}{\partial \theta_2} + \frac{\partial d42}{\partial \theta_2} - \frac{\partial d22}{\partial \theta_4}\right\} = 0$$

$$\boxed{\begin{aligned} C_{\alpha\beta\gamma} &= C_{\beta\alpha\gamma} \\ C_{\alpha\beta\gamma} &= -C_{\alpha\gamma\beta} \end{aligned}}$$

$$c_{242} = -c_{224} = 0 \qquad c_{422} = c_{242} = 0$$

$$c_{233} = \frac{1}{2}\left\{\frac{\partial d33}{\partial \theta_2} + \frac{\partial d32}{\partial d_3} - \frac{\partial d23}{\partial d_3}\right\} = 0$$

$$c_{323} = c_{233} = 0 \qquad c_{332} = -c_{323} = 0$$

$$c_{234} = \frac{1}{2}\left\{\frac{\partial d43}{\partial \theta_2} + \frac{\partial d42}{\partial d_3} - \frac{\partial d23}{\partial \theta_4}\right\} = 0$$

$$c_{324} = c_{234} = 0 \qquad c_{243} = -c_{234} = 0$$

$$c_{423} = c_{243} = 0 \qquad c_{342} = -c_{324} = 0 \qquad c_{432} = c_{342} = 0$$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$C = \begin{bmatrix} c_{111} & c_{121} & c_{131} & c_{141} \\ c_{112} & c_{122} & c_{132} & c_{142} \\ c_{113} & c_{123} & c_{133} & c_{143} \\ c_{114} & c_{124} & c_{134} & c_{144} \end{bmatrix}\dot{\theta}_1 + \begin{bmatrix} c_{211} & c_{221} & c_{231} & c_{241} \\ c_{212} & c_{222} & c_{232} & c_{242} \\ c_{213} & c_{223} & c_{233} & c_{243} \\ c_{214} & c_{224} & c_{234} & c_{244} \end{bmatrix}\dot{\theta}_2$$

$$+ \begin{bmatrix} c_{311} & c_{321} & c_{331} & c_{341} \\ c_{312} & c_{322} & c_{332} & c_{342} \\ c_{313} & c_{323} & c_{333} & c_{343} \\ c_{314} & c_{324} & c_{334} & c_{344} \end{bmatrix}\dot{d}_3 + \begin{bmatrix} c_{411} & c_{421} & c_{431} & c_{441} \\ c_{412} & c_{422} & c_{432} & c_{442} \\ c_{413} & c_{423} & c_{433} & c_{443} \\ c_{414} & c_{424} & c_{434} & c_{444} \end{bmatrix}\dot{\theta}_4$$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$\boxed{\begin{aligned} C_{\alpha\beta\gamma} &= C_{\beta\alpha\gamma} \\ C_{\alpha\beta\gamma} &= -C_{\alpha\gamma\beta} \end{aligned}}$$

$$c_{244} = \frac{1}{2}\left\{\frac{\partial d44}{\partial \theta_2} + \frac{\partial d42}{\partial \theta_4} - \frac{\partial d24}{\partial \theta_4}\right\} = 0$$

$$c_{424} = c_{244} = 0 \qquad c_{442} = -c_{424} = 0$$

$$c_{333} = \frac{1}{2}\left\{\frac{\partial d33}{\partial d_3} + \frac{\partial d33}{\partial d_3} - \frac{\partial d33}{\partial d_3}\right\} = 0$$

$$c_{334} = \frac{1}{2}\left\{\frac{\partial d43}{\partial d_3} + \frac{\partial d43}{\partial d_3} - \frac{\partial d33}{\partial \theta_4}\right\} = 0$$

$$c_{343} = -c_{334} = 0 \qquad c_{433} = -c_{343} = 0$$

$$c_{444} = \frac{1}{2}\left\{\frac{\partial d44}{\partial \theta_4} + \frac{\partial d44}{\partial \theta_4} - \frac{\partial d44}{\partial \theta_4}\right\} = 0$$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$C = \begin{bmatrix} c_{111} & c_{121} & c_{131} & c_{141} \\ c_{112} & c_{122} & c_{132} & c_{142} \\ c_{113} & c_{123} & c_{133} & c_{143} \\ c_{114} & c_{124} & c_{134} & c_{144} \end{bmatrix} \dot{\theta}_1 + \begin{bmatrix} c_{211} & c_{221} & c_{231} & c_{241} \\ c_{212} & c_{222} & c_{232} & c_{242} \\ c_{213} & c_{223} & c_{233} & c_{243} \\ c_{214} & c_{224} & c_{234} & c_{244} \end{bmatrix} \dot{\theta}_2$$

$$+ \begin{bmatrix} c_{311} & c_{321} & c_{331} & c_{341} \\ c_{312} & c_{322} & c_{332} & c_{342} \\ c_{313} & c_{323} & c_{333} & c_{343} \\ c_{314} & c_{324} & c_{334} & c_{344} \end{bmatrix} \dot{d}_3 + \begin{bmatrix} c_{411} & c_{421} & c_{431} & c_{441} \\ c_{412} & c_{422} & c_{432} & c_{442} \\ c_{413} & c_{423} & c_{433} & c_{443} \\ c_{414} & c_{424} & c_{434} & c_{444} \end{bmatrix} \dot{\theta}_4$$

SÍMBOLO DE CHRISTOFFEL

$$C_{ijk} = \frac{1}{2}\left(\frac{\partial d_{kj}}{\partial q_i} + \frac{\partial d_{ki}}{\partial q_j} - \frac{\partial d_{ij}}{\partial q_k}\right)$$

$$c_{344} = \frac{1}{2}\left\{\frac{\partial d43}{\partial d_3} + \frac{\partial d43}{\partial d_3} - \frac{\partial d33}{\partial \theta_4}\right\} = 0$$

$$\boxed{\begin{aligned} C_{\alpha\beta\gamma} &= C_{\beta\alpha\gamma} \\ C_{\alpha\beta\gamma} &= -C_{\alpha\gamma\beta} \end{aligned}}$$

$$c_{434} = c_{344} = 0 \qquad c_{443} = -c_{434} = 0$$

$$C = \begin{bmatrix} c_{111} & c_{121} & c_{131} & c_{141} \\ c_{112} & c_{122} & c_{132} & c_{142} \\ c_{113} & c_{123} & c_{133} & c_{143} \\ c_{114} & c_{124} & c_{134} & c_{144} \end{bmatrix}\dot{\theta}_1 + \begin{bmatrix} c_{211} & c_{221} & c_{231} & c_{241} \\ c_{212} & c_{222} & c_{232} & c_{242} \\ c_{213} & c_{223} & c_{233} & c_{243} \\ c_{214} & c_{224} & c_{234} & c_{244} \end{bmatrix}\dot{\theta}_2$$

$$+ \begin{bmatrix} c_{311} & c_{321} & c_{331} & c_{341} \\ c_{312} & c_{322} & c_{332} & c_{342} \\ c_{313} & c_{323} & c_{333} & c_{343} \\ c_{314} & c_{324} & c_{334} & c_{344} \end{bmatrix}\dot{d}_3 + \begin{bmatrix} c_{411} & c_{421} & c_{431} & c_{441} \\ c_{412} & c_{422} & c_{432} & c_{442} \\ c_{413} & c_{423} & c_{433} & c_{443} \\ c_{414} & c_{424} & c_{434} & c_{444} \end{bmatrix}\dot{\theta}_4$$

SÍMBOLO DE CHRISTOFFEL

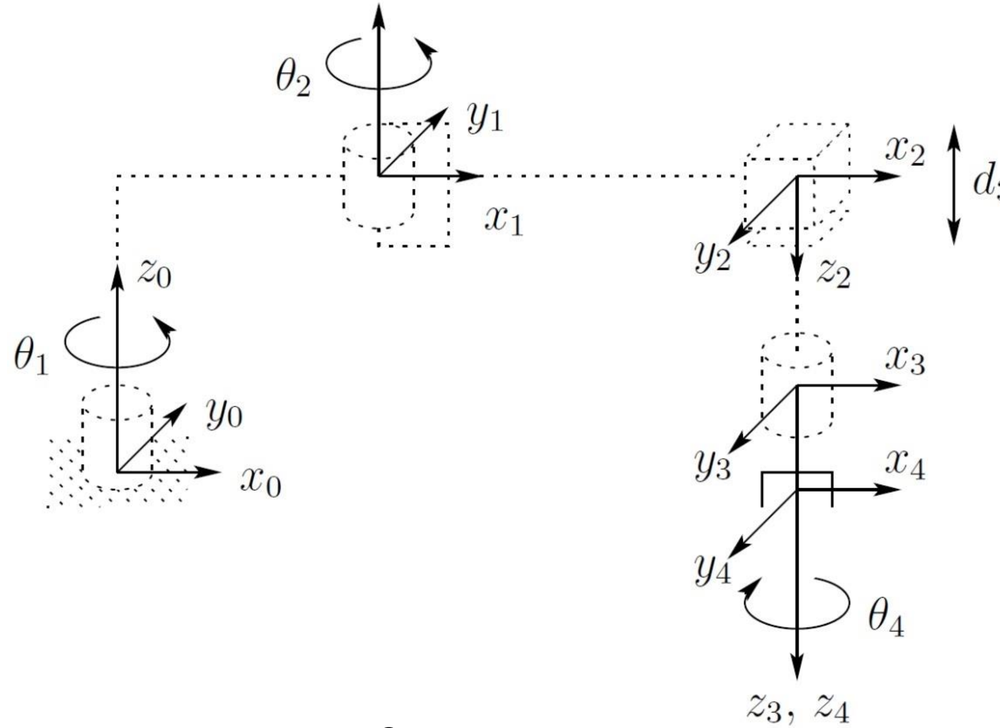
$$C = \begin{bmatrix} c_{111} & c_{121} & c_{131} & c_{141} \\ c_{112} & c_{122} & c_{132} & c_{142} \\ c_{113} & c_{123} & c_{133} & c_{143} \\ c_{114} & c_{124} & c_{134} & c_{144} \end{bmatrix} \dot{\theta}_1 + \begin{bmatrix} c_{211} & c_{221} & c_{231} & c_{241} \\ c_{212} & c_{222} & c_{232} & c_{242} \\ c_{213} & c_{223} & c_{233} & c_{243} \\ c_{214} & c_{224} & c_{234} & c_{244} \end{bmatrix} \dot{\theta}_2$$

$$+ \begin{bmatrix} c_{311} & c_{321} & c_{331} & c_{341} \\ c_{312} & c_{322} & c_{332} & c_{342} \\ c_{313} & c_{323} & c_{333} & c_{343} \\ c_{314} & c_{324} & c_{334} & c_{344} \end{bmatrix} \dot{d}_3 + \begin{bmatrix} c_{411} & c_{421} & c_{431} & c_{441} \\ c_{412} & c_{422} & c_{432} & c_{442} \\ c_{413} & c_{423} & c_{433} & c_{443} \\ c_{414} & c_{424} & c_{434} & c_{444} \end{bmatrix} \dot{\theta}_4$$

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & C_{34} \\ C_{41} & C_{42} & C_{43} & C_{44} \end{bmatrix}$$

$$C = \begin{bmatrix} -\dot{\theta}_2 a_1 a_2 S_2 (m_2 + m_3 + m_4) & -\dot{\theta}_1 a_1 a_2 S_2 (m_2 + m_3 + m_4) & 0 & 0 \\ \dot{\theta}_1 a_1 a_2 S_2 (m_2 + m_3 + m_4) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

ENERGIA POTENCIAL



$$P_1 = m_1 g d_1$$
$$P_2 = m_2 g d_1$$
$$P_3 = m_3 g(d_1 - d_3)$$
$$P_4 = m_4 g(d_1 - d_3 - d_4)$$

$$P = P_1 + P_2 + P_3 + P_4$$

$$g_1 = \frac{\partial P}{\partial \theta_1} = 0 \qquad g_2 = \frac{\partial P}{\partial \theta_2} = 0 \qquad g_3 = \frac{\partial P}{\partial d_3} = -g(m_3 + m_4)$$

$$g_4 = \frac{\partial P}{\partial \theta_4} = 0 \qquad g = \begin{bmatrix} g_1 \\ g_2 \\ g_3 \\ g_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -g(m_3 + m_4) \\ 0 \end{bmatrix}$$

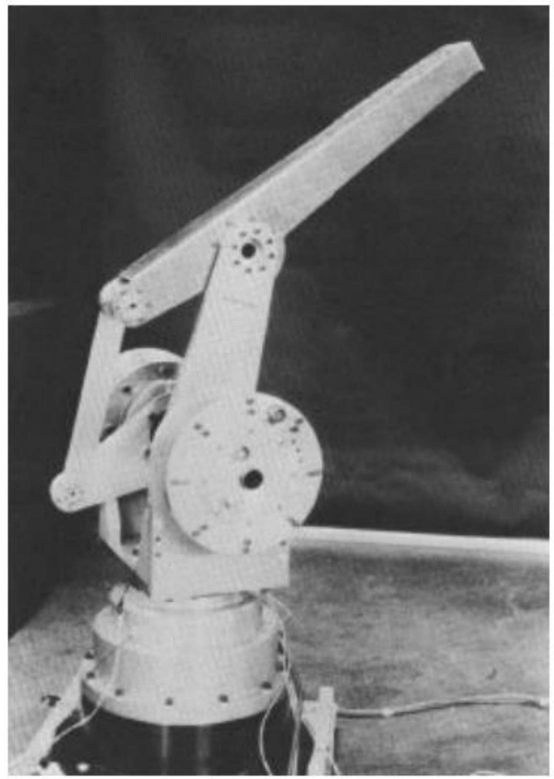
EQUAÇÃO DE MOVIMENTO

$$D(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = \tau$$

$$\begin{bmatrix} d_{11} & d_{12} & d_{13} & d_{14} \\ d_{21} & d_{22} & d_{23} & d_{24} \\ d_{31} & d_{32} & d_{33} & d_{34} \\ d_{41} & d_{42} & d_{43} & d_{44} \end{bmatrix} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \\ \ddot{d}_3 \\ \ddot{\theta}_4 \end{Bmatrix} + \begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & C_{34} \\ C_{41} & C_{42} & C_{43} & C_{44} \end{bmatrix} \begin{Bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{d}_3 \\ \dot{\theta}_4 \end{Bmatrix}$$

$$+ \begin{bmatrix} 0 \\ 0 \\ -g(m_3 + m_4) \\ 0 \end{bmatrix} = \begin{Bmatrix} \tau_1 \\ \tau_2 \\ F_3 \\ \tau_4 \end{Bmatrix}$$

EX: 5 BARRAS



Comau Smart NJ130

MIT Direct Drive Mark II and Mark III

EX: 5 BARRAS

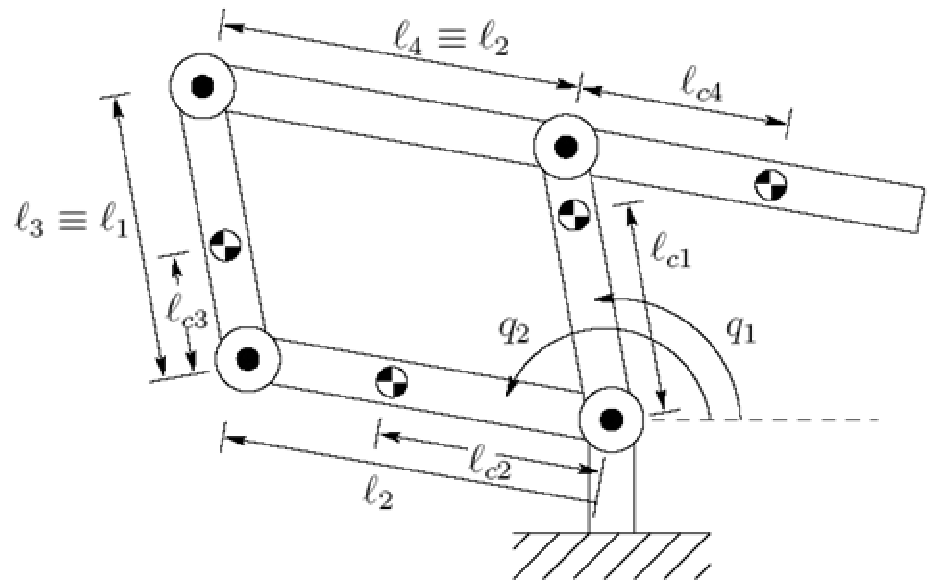
Coordenadas do centro de massa:

$$\begin{bmatrix} x_{c1} \\ y_{c1} \end{bmatrix} = \begin{bmatrix} L_{c1}\cos q_1 \\ L_{c1}\sin q_1 \end{bmatrix}$$

$$\begin{bmatrix} x_{c2} \\ y_{c2} \end{bmatrix} = \begin{bmatrix} L_{c2}\cos q_2 \\ L_{c2}\sin q_2 \end{bmatrix}$$

$$\begin{bmatrix} x_{c3} \\ y_{c3} \end{bmatrix} = \begin{bmatrix} L_2\cos q_2 \\ L_2\sin q_2 \end{bmatrix} + \begin{bmatrix} L_{c3}\cos q_1 \\ L_{c3}\sin q_1 \end{bmatrix}$$

$$\begin{bmatrix} x_{c4} \\ y_{c4} \end{bmatrix} = \begin{bmatrix} L_1\cos q_1 \\ L_1\sin q_1 \end{bmatrix} - \begin{bmatrix} L_{c4}\cos q_2 \\ L_{c4}\sin q_2 \end{bmatrix}$$



EX: 5 BARRAS

- Velocidades lineares:

$$v_{c1} = \begin{bmatrix} -L_{c1}\sin q_1 & 0 \\ L_{c1}\cos q_1 & 0 \\ 0 & 0 \end{bmatrix}\dot{q},\ v_{c2} = \begin{bmatrix} 0 & -L_{c2}\sin q_2 \\ 0 & L_{c2}\cos q_2 \\ 0 & 0 \end{bmatrix}\dot{q}$$

$$v_{c3} = \begin{bmatrix} -L_{c3}\sin q_1 & -L_2\sin q_2 \\ L_{c3}\cos q_1 & L_2\cos q_2 \\ 0 & 0 \end{bmatrix}\dot{q},\ v_{c4} = \begin{bmatrix} -L_1\sin q_1 & -L_{c4}\sin q_2 \\ L_1\cos q_1 & L_{c4}\cos q_2 \\ 0 & 0 \end{bmatrix}\dot{q}$$

- Velocidades angulares:

$$\omega_1 = \omega_3 = \dot{q}_1\hat{k}$$

$$\omega_2 = \omega_4 = \dot{q}_2\hat{k}$$

EX: 5 BARRAS

- Matriz de inércia:

$$D(q) = \sum_{i=1}^{4} m_i J_{v_i}(q)^T J_{v_i}(q) + \begin{bmatrix} I_1 + I_3 & 0 \\ 0 & I_2 + I_4 \end{bmatrix}$$

$$= \begin{bmatrix} m_1 L_{c1}^{\ 2} + m_3 L_{c3}^{\ 2} + m_4 L_1^{\ 2} + I_1 + I_3 & (m_3 L_2 L_{c3} - m_4 L_1 L_{c4})\cos(q_2 - q_1) \\ (m_3 L_2 L_{c3} - m_4 L_1 L_{c4})\cos(q_2 - q_1) & m_2 L_{c2}^{\ 2} + m_3 L_2^{\ 2} + m_4 L_{c4}^{\ 2} + I_2 + I_4 \end{bmatrix}$$

$$m_3 L_2 L_{c3} = m_4 L_1 L_{c4}$$

- Fazendo algumas suposições:
  - A matriz de inercia é diagonal e constante
  - As derivadas parciais e os símbolos de Christoffel são zero

EX: 5 BARRAS

- A energia potencial:

$$P = \sum_{i=1}^{4} P_i = g\sum_{i=1}^{4} m_i y_i$$
$$= gm_1L_{c1}\sin q_1 + gm_2L_{c2}\sin q_2$$
$$+ gm_3\left(L_2\sin q_2 + L_{c3}\sin q_1\right) + gm_4\left(L_1\sin q_1 - L_{c4}\sin q_2\right)$$

- A gravidade:

$$g_1 = g\cos q_1\left(m_1L_{c1} + m_3L_{c3} + m_4L_1\right)$$
$$g_2 = g\cos q_2\left(m_2L_{c2} + m_3L_2 - m_4L_{c4}\right)$$

- Equações de movimento:

$$d_{11}\ddot{q}_1 + g_1(q_1) = \tau_1$$
$$d_{22}\ddot{q}_2 + g_2(q_2) = \tau_2$$

FORMULAÇÃO DE NEWTON-EULER

- A formulação de Newton-Euler leva aos mesmos resultados obtidos através da formulação de Euler-Lagrange, embora percorrendo um caminho bastante diferente. Na formulação Lagrangiana, trata-se o robô como um todo e realiza-se a análise usando a função Lagrangiana (diferença entre as energias cinética e potencial). Na formulação Newtoniana, considera-se cada membro separadamente e escreve-se as equações que descrevem os seus movimentos linear e angular. Evidentemente, como cada membro está acoplado aos demais, a equação do movimento de um membro contem forças e torques de restrição, que aparecem também nas equações do movimento dos demais membros. Utilizando um procedimento recursivo, é possível determinar todos esses termos de acoplamento e eventualmente chegar à descrição do manipulador como um todo.

FORMULAÇÃO DE NEWTON-EULER

- A Mecânica Newtoniana repousa sobre as seguintes Leis:
- 1. **Terceira Lei de** Newton:
- "*A toda ação corresponde uma reação igual e de sentido oposto*".
- Assim, se o corpo 1 aplica uma força **f** e um torque t sobre o corpo 2, então o corpo 2 aplica uma força -**f** e um torque -t sobre o corpo 1.
- 2. **Segunda Lei de Newton**:
- "*A taxa de variação da quantidade de movimento linear é igual à força total aplicada ao corpo*".

$$\sum f_i = \frac{d}{dt}(mv_c) = m\dot{v}_c$$

FORMULAÇÃO DE NEWTON-EULER

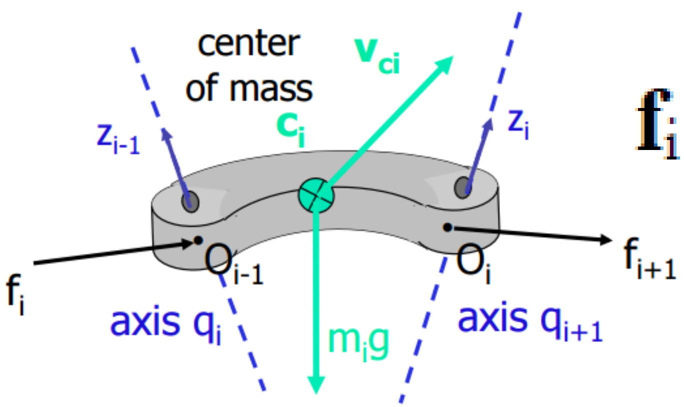
- 3. **Lei de Euler**:
- "*A taxa de variação da quantidade de movimento angular é igual ao torque total aplicado ao corpo*".

$$\sum \mu_i = \frac{d}{dt}(I\omega) = I\dot{\omega} + \frac{d}{dt}\left(R\bar{I}R^T\right)\omega = I\dot{\omega} + \left(\dot{R}\bar{I}R^T + R\bar{I}\dot{R}^T\right)\omega$$

$$= I\dot{\omega} + S(\omega)R\bar{I}R^T\omega + R\bar{I}R^TS^T\cancel{(\omega)\omega} = I\dot{\omega} + \omega \times I\omega$$

FORMULAÇÃO DE NEWTON-EULER

- **fi** é a força aplicada pelo corpo i-1 sobre o corpo i. Pela 3ª Lei de Newton, o corpo i+1 aplica uma força $-\mathbf{f}_{i+1}$ sobre o corpo i, devendo-se pré-multiplicar essa força pela matriz de rotação $\mathbf{R}^{i}{}_{i+1}$. obtendo-se $-\mathbf{R}^{i+1}{}_{i}\ \mathbf{f}_{i+1}$. Explanação análoga se aplica aos torques ti e $-\mathbf{R}^{i+1}{}_{i}$ ti+1. A força $m_i\mathbf{g}_i$ é a força gravitacional. Escrevendo a 2ª Lei de Newton para o corpo i:

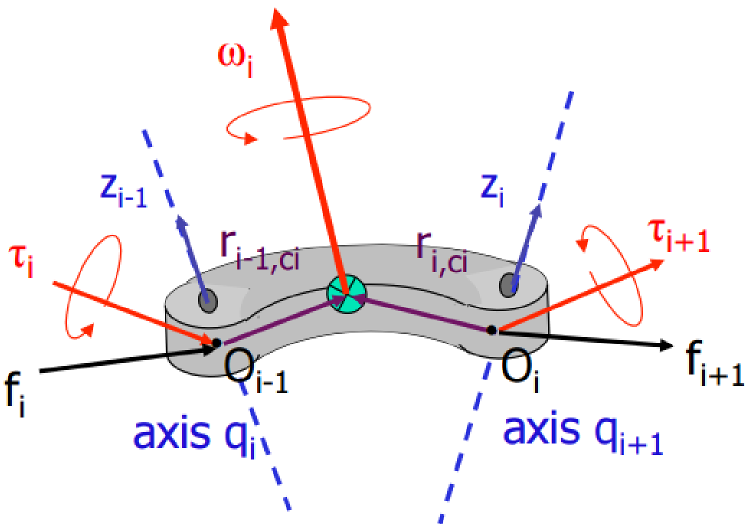


$$\mathbf{f}_i - \mathbf{R}^{i+1}{}_{i}\ \mathbf{f}_{i+1} + m_i\mathbf{g}_i = m_i\mathbf{a}_{c,i}$$

FORMULAÇÃO DE NEWTON-EULER

- O momento exercido por uma força **f** em torno de um ponto P pode ser dado por **f** x **r**, onde **r** *é o vetor cuja origem é o ponto de aplicação da força e cuja extremidade é o ponto P*. Além disso, como o peso não exerce momento em relação ao centro de massa, tem-se a Lei de Euler.

$$\tau_i - \mathbf{R}^{i+1}{}_i\, \tau_{i+1} + \mathbf{f}_i \times \mathbf{r}_{i,ci} - (\mathbf{R}^{i+1}{}_i \mathbf{f}_{i+1}) \times \mathbf{r}_{i+1,ci} = \mathrm{I}_i\, \alpha_i + \omega_i \times (\mathrm{I}_i\, \omega_i)$$



FORMULAÇÃO DE NEWTON-EULER

- A essência da formulação de Newton-Euler consiste em achar as forças e os torques que correspondem a um dado conjunto de coordenadas generalizadas e suas duas primeiras derivadas temporais. Essa informação pode ser usada tanto para obter equações em forma fechada que descrevem a evolução temporal das coordenadas generalizadas, como para saber quais forças generalizadas necessitam ser aplicadas, a fim de realizar uma trajetória particular.
- A ideia geral pode ser estabelecida assim: dados $\ddot{q}, \dot{q}$ e q, suponha-se que, de algum modo, seja possível determinar todas as velocidades e acelerações de várias partes do manipulador, isto é, todas as quantidades $\boldsymbol{a}_{c,i}$, w_i e a_i.

$$\mathbf{f}_i = \mathbf{R}^{i+1}{}_i\, \mathbf{f}_{i+1} + m_i \mathbf{a}_{c,i} - m_i \mathbf{g}_i \qquad \left\{ \begin{array}{l} \mathbf{f}_{n+1} = 0 \\ \tau_{n+1} = 0 \end{array} \right.$$

FORMULAÇÃO DE NEWTON-EULER

$$\tau_i = \mathbf{R}^{i+1}{}_i\, \tau_{i+1} - \mathbf{f}_i \times \mathbf{r}_{i,ci} + (\mathbf{R}^{i+1}{}_i \mathbf{f}_{i+1}) \times \mathbf{r}_{i+1,ci} + I_i\, \alpha_i + \omega_i \times (I_i\, \omega_i)$$

- A solução estará completa calculando-se as relações entre $\ddot{q}, \dot{q}$ e q e $\mathbf{a}_{c,i}$, ω_i e α_i. Isso pode ser feito através de um procedimento recursivo no sentido crescente de i.

❑**Caso de juntas rotativas**

Velocidade angular do sistema i: $\omega_i^{(0)} = \omega_{i-1}^{(0)} + z_{i-1}\dot{q}_i$

Para obter uma relação entre ω_i e ω_{i-1} e preciso apenas expressar a equação acima no sistema i:

$$\omega_i = \left(R_i^{i-1}\right)^T \omega_{1-1} + b_i\dot{q}_i \quad \Rightarrow \quad b_i = \left(R_i^0\right)^T z_{i-1}$$

FORMULAÇÃO DE NEWTON-EULER

- Aceleração angular: $\alpha_i = \left(R_i^0\right)^T \dot{\omega}_i^{(0)}$

- Velocidade e a aceleração angular no centro de massa:

$$\dot{\omega}_i^{(0)} = \dot{\omega}_{i-1}^{(0)} + z_{i-1}\ddot{q}_i + \omega_i^{(0)} \times z_{i-1}\dot{q}_i$$

$$\alpha_i = \left(R_i^{i-1}\right)^T \alpha_{i-1} + b_i\ddot{q}_i + \omega_i \times b_i\dot{q}_i$$

- Velocidade e a aceleração linear:

$$v_{c,i}^{(0)} = v_{e,i-1}^{(0)} + \omega_i^{(0)} \times r_{i,ci}^{(0)}$$

$$a_{c,i}^{(0)} = a_{e,i-1}^{(0)} \times r_{i,ci}^{(0)} + \omega_i^{(0)} \times (\omega_i^{(0)} \times r_{i,ci}^{(0)})$$

FORMULAÇÃO DE NEWTON-EULER

- Transformando $\mathbf{a}_{c,i-1}$ para o sistema i:

$$a_{c,i} = \left(R_i^{i-1}\right)^T a_{e,i-1} + \dot{\omega}_i \times r_{i,ci} + \omega_i \times \left(\omega_i \times r_{i,ci}\right)$$

$$R_{i-1}^i = \left(R_i^{i-1}\right)^T$$

$$a_{c,i} = \left(R_0^i\right) a_{c,i}^{(0)}$$

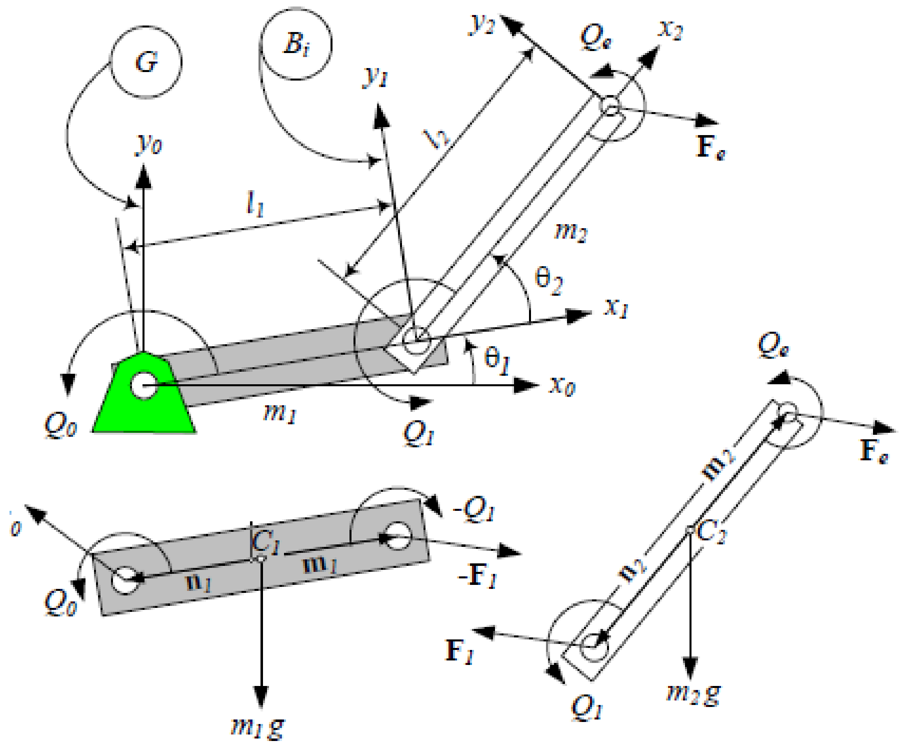
$$R(a \times b) = (Ra) \times (Rb)$$

- Aceleração no final do elo i:

$$a_{e,i} = \left(R_i^{i-1}\right)^T a_{e,i-1} + \dot{\omega}_i \times r_{i,i+1} + \omega_i \times \left(\omega_i \times r_{i,i+1}\right)$$

EX.: MANIPULADOR PLANAR

- Equações de movimento aplicando a formulação recursiva de Newton-Euler.



EX.: MANIPULADOR PLANAR

- Velocidades e acelerações das juntas é definido por:

$$\omega_1 = \begin{bmatrix} 0 \\ 0 \\ \dot{q}_1 \end{bmatrix} = \dot{q}_1.\hat{k} \qquad \alpha_1 = \begin{bmatrix} 0 \\ 0 \\ \ddot{q}_1 \end{bmatrix} = \ddot{q}_1.\hat{k}$$

$$\omega_2 = \begin{bmatrix} 0 \\ 0 \\ \dot{q}_1 + \dot{q}_2 \end{bmatrix} = (\dot{q}_1+\dot{q}_2).\hat{k} \qquad \alpha_2 = \begin{bmatrix} 0 \\ 0 \\ \ddot{q}_1 + \ddot{q}_2 \end{bmatrix} = (\ddot{q}_1+\ddot{q}_2).\hat{k}$$

- Aceleração no final do elo i:

$$\begin{aligned} r_{1,c1} &= \ell_{c1}i,\ r_{2,c1} = (\ell_1 - \ell_{c1})i,\ r_{1,2} = \ell_1 i \\ r_{2,c2} &= \ell_{c2}i,\ r_{3,c2} = (\ell_2 - \ell_{c2})i,\ r_{2,3} = \ell_2 i \end{aligned}$$

EX.: MANIPULADOR PLANAR

- Avanço Elo 1
  - Aplicando a equação de aceleração do centro de massa para o elo 1:
    - Para o caso do primeiro Elo: $$a_{e,0} = 0$$
    - Então:

$$\begin{aligned} a_{c,1} &= \ddot{q}_1 k \times \ell_{c1} i + \dot{q}_1 k \times (\dot{q}_1 k \times \ell_{c1} i) \\ &= \ell_{c1} \ddot{q}_1 j - \ell_{c1} \dot{q}_1^2 i = \begin{bmatrix} -\ell_{c1} \dot{q}_1^2 \\ \ell_c \ddot{q}_1 \\ 0 \end{bmatrix} \end{aligned}$$

Lembrando...

$$a_{e,i} = \left(R_i^{i-1}\right)^T a_{e,i-1} + \dot{\omega}_i \times r_{i,i+1} + \omega_i \times \left(\omega_i \times r_{i,i+1}\right)$$

EX.: MANIPULADOR PLANAR

- Avanço Elo 1
  - A aceleração da gravidade e aceleração no final do elo i:

$$g_1 = -(R_0^1)^T g j = g \begin{bmatrix} \sin q_1 \\ -\cos q_1 \\ 0 \end{bmatrix}$$

  - Como o robô é planar, as acelerações e torques na direção z são nulas. Para facilitar os cálculos a última linha dos vetores será desconsiderada.

$$a_{e,1} = \begin{bmatrix} -l_1 \dot{q}_1^2 \\ l_1 \ddot{q}_1 \\ \end{bmatrix}$$

EX.: MANIPULADOR PLANAR

- Avanço Elo 2
  - Aceleração do centro de massa

$$\alpha_{c,2} = (R_2^1)^T a_{e,1} + (\ddot{q}_1 + \ddot{q}_2)\hat{k} \times l_{c2}\hat{\imath} + (\dot{q}_1 + \dot{q}_2)\hat{k} \times \left[(\dot{q}_1 + \dot{q}_2)\hat{k} \times l_{c2}\hat{\imath}\right]$$

$$(R_2^1)^T a_{e,1} = \begin{bmatrix} \cos q_2 & \sin q_2 \\ -\sin q_2 & \cos q_2 \end{bmatrix} \begin{bmatrix} -\ell_1 \dot{q}_1^2 \\ \ell_1 \ddot{q}_1 \end{bmatrix} = \begin{bmatrix} -\ell_1 \dot{q}_1^2 \cos q_2 + \ell_1 \ddot{q}_1 \sin \dot{q}_2 \\ \ell_1 \dot{q}_1^2 \sin q_2 + \ell_1 \ddot{q}_1 \cos q_2 \end{bmatrix}$$

$$a_{c,2} = \begin{bmatrix} -\ell_1 \dot{q}_1^2 \cos q_2 + \ell_1 \ddot{q}_1 \sin q_2 - \ell_{c2}(\dot{q}_1 + \dot{q}_2)^2 \\ \ell_1 \dot{q}_1^2 \sin q_2 + \ell_1 \ddot{q}_1 \cos q_2 - \ell_{c2}(\ddot{q}_1 + \ddot{q}_2) \end{bmatrix}$$

  - Aceleração da gravidade

$$g_2 = g \begin{bmatrix} \sin(q_1 + q_2) \\ -\cos(q_1 + q_2) \end{bmatrix}$$

EX.: MANIPULADOR PLANAR

- Retroativo Elo 2

$$\begin{aligned} f_2 &= m_2 a_{c,2} - m_2 g_2 \\ \tau_2 &= I_2 \alpha_2 + \omega_2 \times (I_2 \omega_2) - f_2 \times \ell_{c2} i \end{aligned}$$

- Substituindo:

$$\begin{aligned} \tau_2 = \;& I_2(\ddot{q}_1 + \ddot{q}_2)k + [m_2 \ell_1 \ell_{c2} \sin q_2 \dot{q}_1^2 + m_2 \ell_1 \ell_{c2} \cos q_2 \ddot{q}_1 \\ & + m_2 \ell_{c2}^2 (\ddot{q}_1 + \ddot{q}_2) + m) 2 \ell_{c2} g \cos(q_1 + q_2)]k \end{aligned}$$

EX.: MANIPULADOR PLANAR

- Retroativo Elo 1

$$\begin{aligned} f_1 &= m_1 a_{c,1} + R_1^2 f_2 - m_1 g_1 \\ \tau_1 &= R_1^2 \tau_2 - f_1 \times \ell_{c,1} i - (R_1^2 f_2) \times (\ell_1 - \ell_{c1}) i \\ &\quad + I_1 \alpha_1 + \omega_1 \times (I_1 \omega_1) \end{aligned}$$

- Substituindo:

$$\begin{aligned} \tau_1 &= \tau_2 + m_1 \ell_{c1}^2 + m_1 \ell_{c1} g \cos q_1 + m_2 \ell_1 g \cos q_1 + I_1 \ddot{q}_1 \\ &\quad + m_2 \ell_1^2 \ddot{q}_1 - m_1 \ell_1 \ell_{c2} (\dot{q}_1 + \dot{q}_2)^2 \sin q_2 + m_2 \ell_1 \ell_{c2} (\ddot{q}_1 + \ddot{q}_2) \cos q_2 \end{aligned}$$

BIBLIOGRAFIA BÁSICA UTILIZADA

- SPONG M.W. Hutchinson, S., and Vidyasagar, M., "Robot Modeling and Control", John Wiley and Sons, 2006.
- Corke, Peter. Robotics, vision and control: fundamental algorithms in MATLAB. Vol. 73. Springer, 2011.
- Siciliano, Bruno, et al. *Robotics: modelling, planning and control*. Springer Science & Business Media, 2010.
- Jazar, Reza N. *Theory of applied robotics: kinematics, dynamics, and control*. Springer Science & Business Media, 2010.
- Siciliano and Oussama Khatib, eds. Springer handbook of robotics. Springer Science & Business Media, 2008
- Notas de Aulas – Prof. Marcelo Becker
- Notas de Aulas - Prof. Alessandro De Luca